CYBERSTER INTERNSHIP PROGRAM

WEEK 1
REPORT

Reconnaissance & Information Gathering

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INSTITUTION
CYBERSTER

REPORT PERIOD
Week 1

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Executive Summary

Week 1 of the Cyberster internship focused on fundamental reconnaissance and information gathering techniques. Through comprehensive hands-on exercises, this report documents successful execution of four major reconnaissance tasks targeting hackthissite.com and hackthissite.org domains.

The objectives included discovering organizational digital assets through passive reconnaissance, performing DNS enumeration to map infrastructure, conducting open-source intelligence (OSINT) on public repositories, and analyzing version control systems for intelligence. All tasks were completed successfully with detailed documentation of methodologies, tools utilized, and findings.

Key finding: publicly available information reveals substantial organizational infrastructure details. This training established foundational knowledge for subsequent weeks' advanced security assessments.

Learning Objectives

Upon completion of Week 1, the following objectives were achieved:

• Understand reconnaissance role in cybersecurity attack chain and defensive operations

• Master passive reconnaissance techniques that gather intelligence without triggering alerts

• Effectively utilize multiple subdomain enumeration tools and consolidate results

• Perform comprehensive DNS enumeration to map organizational infrastructure

• Conduct OSINT investigations on public repositories to identify exposed data

• Analyze version control systems for development history and security insights

• Implement defensive measures to protect assets against reconnaissance

Environment Setup and Tools

Operating System: Kali Linux 2026.2 (Debian-based, specialized for penetration testing)

Virtualization: VirtualBox 7.0 with Linux kernel modules

VM Configuration: 4 cores CPU | 8 GB RAM | 60 GB Disk | Bridged Network

Installed Tools:

Tool	Version	Purpose
Subfinder	v2.5.5	Multi-source subdomain discovery
Assetfinder	v0.1.1	Certificate transparency logs
Findomain	v8.1+	Cross-source enumeration
DIG	v9.16+	DNS enumeration and queries
cURL	v7.68+	API interaction and data transfer

Task 1: Subdomain Enumeration Using OSINT Tools

Objective

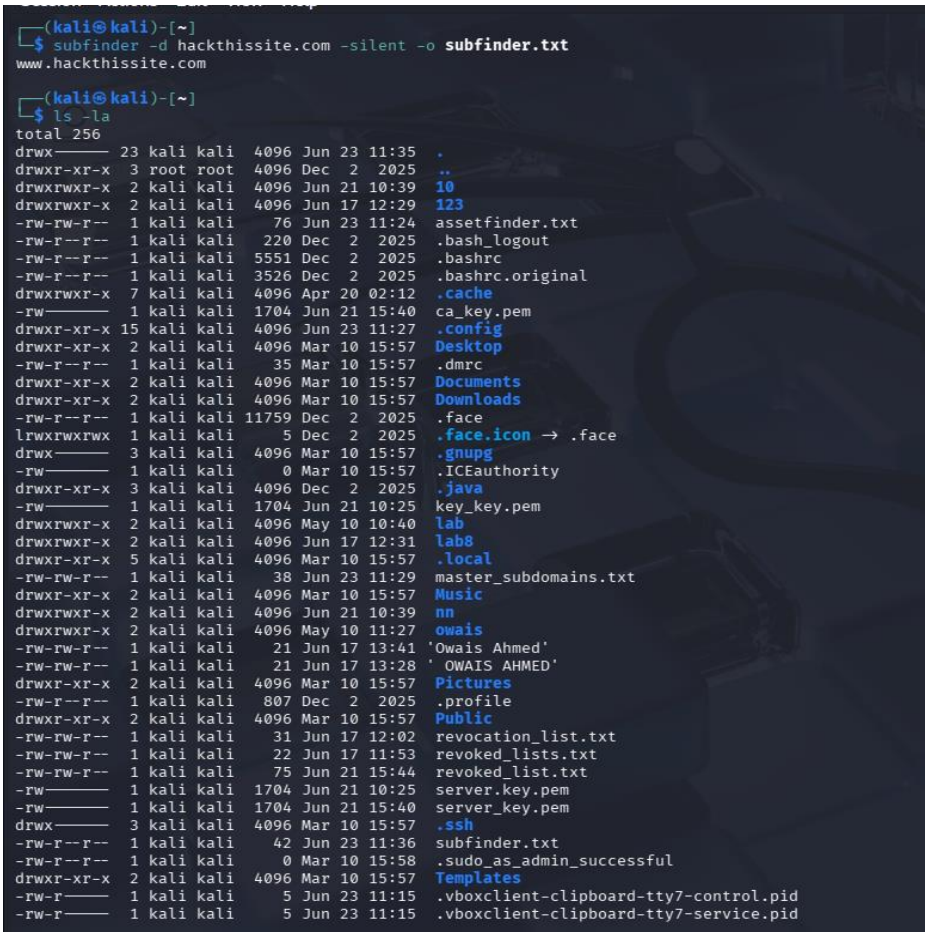
Discover and enumerate all subdomains associated with hackthissite.com through passive reconnaissance techniques using industry-standard OSINT tools. This task demonstrates how attackers map organizational digital assets without triggering defensive monitoring systems.

Theory and Background

Subdomain enumeration identifies all subdomains under a target domain. Organizations deploy services across numerous subdomains including web applications, APIs, mail servers, and development environments. Modern OSINT leverages publicly available data sources including SSL certificate transparency logs, DNS query history, and web crawler data. These sources require no direct queries to target infrastructure, making enumeration completely passive and undetectable.

Tools: Subfinder (20+ sources), Assetfinder (CT logs), Findomain (cross-source validation)

Task 1 Screenshots



```
drwxr-xr-x  2 kali kali  4096 Mar 10 15:57  Music
drwxrwxr-x  2 kali kali  4096 Jun 21 10:39  nn
drwxrwxr-x  2 kali kali  4096 May 10 11:27  owais
-rw-rw-r--  1 kali kali    21 Jun 17 13:41 'Owais Ahmed'
-rw-rw-r--  1 kali kali    21 Jun 17 13:28 ' OWAIS AHMED'
drwxr-xr-x  2 kali kali  4096 Mar 10 15:57  Pictures
-rw-r--r--  1 kali kali   807 Dec  2  2025  .profile
drwxr-xr-x  2 kali kali  4096 Mar 10 15:57  Public
-rw-rw-r--  1 kali kali    31 Jun 17 12:02  revocation_list.txt
-rw-rw-r--  1 kali kali    22 Jun 17 11:53  revoked_lists.txt
-rw-rw-r--  1 kali kali    75 Jun 21 15:44  revoked_list.txt
-rw-------  1 kali kali  1704 Jun 21 10:25  server.key.pem
-rw-------  1 kali kali  1704 Jun 21 15:40  server_key.pem
drwx------  3 kali kali  4096 Mar 10 15:57  .ssh
-rw-r--r--  1 kali kali    42 Jun 23 11:36  subfinder.txt
-rw-r--r--  1 kali kali     0 Mar 10 15:58  .sudo_as_admin_successful
drwxr-xr-x  2 kali kali  4096 Mar 10 15:57  Templates
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-clipboard-tty7-control.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-clipboard-tty7-service.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-draganddrop-tty7-control.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-draganddrop-tty7-service.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-hostversion-tty7-control.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-seamless-tty7-control.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-seamless-tty7-service.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-vmsvga-session-tty7-control.pid
-rw-r-----  1 kali kali     5 Jun 23 11:15  .vboxclient-vmsvga-session-tty7-service.pid
drwxr-xr-x  2 kali kali  4096 Mar 10 15:57  Videos
drwxrwxr-x  2 kali kali  4096 Jun 23 11:16  week1_recon
-rw-------  1 kali kali    49 Jun 23 11:15  .Xauthority
-rw-------  1 kali kali  4303 Jun 23 11:15  .xsession-errors
-rw-------  1 kali kali  4303 Jun 21 15:39  .xsession-errors.old
-rw-r--r--  1 kali kali   336 Dec  2  2025  .zprofile
-rw-------  1 kali kali  6178 Jun 23 11:35  .zsh_history
-rw-r--r--  1 kali kali 10855 Dec  2  2025  .zshrc

┌──(kali㉿kali)-[~]
└─$ cat subfinder.txt
www.hackthissite.com
www.hackthissite.com

┌──(kali㉿kali)-[~]
└─$ cat subfinder.txt | head -20
www.hackthissite.com
www.hackthissite.com

┌──(kali㉿kali)-[~]
└─$ wc -l subfinder.txt
2 subfinder.txt
```

```
┌──(kali㉿kali)-[~]
└─$ assetfinder --subs-only hackthissite.com > assetfinder.txt

┌──(kali㉿kali)-[~]
└─$ ls -la
total 256
drwx------ 23 kali kali  4096 Jun 23 11:39  .
drwxr-xr-x  3 root root  4096 Dec  2  2025  ..
drwxrwxr-x  2 kali kali  4096 Jun 21 10:39  10
drwxrwxr-x  2 kali kali  4096 Jun 17 12:29  123
-rw-rw-r--  1 kali kali    38 Jun 23 11:39  assetfinder.txt
```

```
┌──(kali㉿kali)-[~]
└─$ cat assetfinder.txt | head -20
hackthissite.com
www.hackthissite.com

┌──(kali㉿kali)-[~]
└─$ wc -l assetfinder.txt
2 assetfinder.txt
```

```
┌──(kali㉿kali)-[~]
└─$ findomain -t hackthissite.com -u findomain.txt

Target ⟹ hackthissite.com

Searching in the CertSpotter API ... 🔍
Searching in the Sublist3r API ... 🔍
Searching in the Crtsh database API ... 🔍
Searching in the Threatcrowd API ... 🔍
Searching in the AnubisDB API ... 🔍
Searching in the Urlscan.io API ... 🔍
Searching in the Threatminer API ... 🔍


hackthissite.com
www.hackthissite.com

Job finished in 2 seconds.
>> 📁 Subdomains for hackthissite.com were saved in: ./findomain.txt 😀

Good luck Hax0r 💀 !


┌──(kali㉿kali)-[~]
└─$ cat findomain.txt | head -20
hackthissite.com
www.hackthissite.com

┌──(kali㉿kali)-[~]
└─$ wc -l findomain.txt
2 findomain.txt
```

```
┌──(kali㉿kali)-[~]
└─$ cat subfinder.txt assetfinder.txt findomain.txt | sort -u > master_subdomains.txt

┌──(kali㉿kali)-[~]
└─$ wc -l master_subdomains.txt
2 master_subdomains.txt

┌──(kali㉿kali)-[~]
└─$ cat master_subdomains.txt | head -20
hackthissite.com
www.hackthissite.com

┌──(kali㉿kali)-[~]
└─$
```

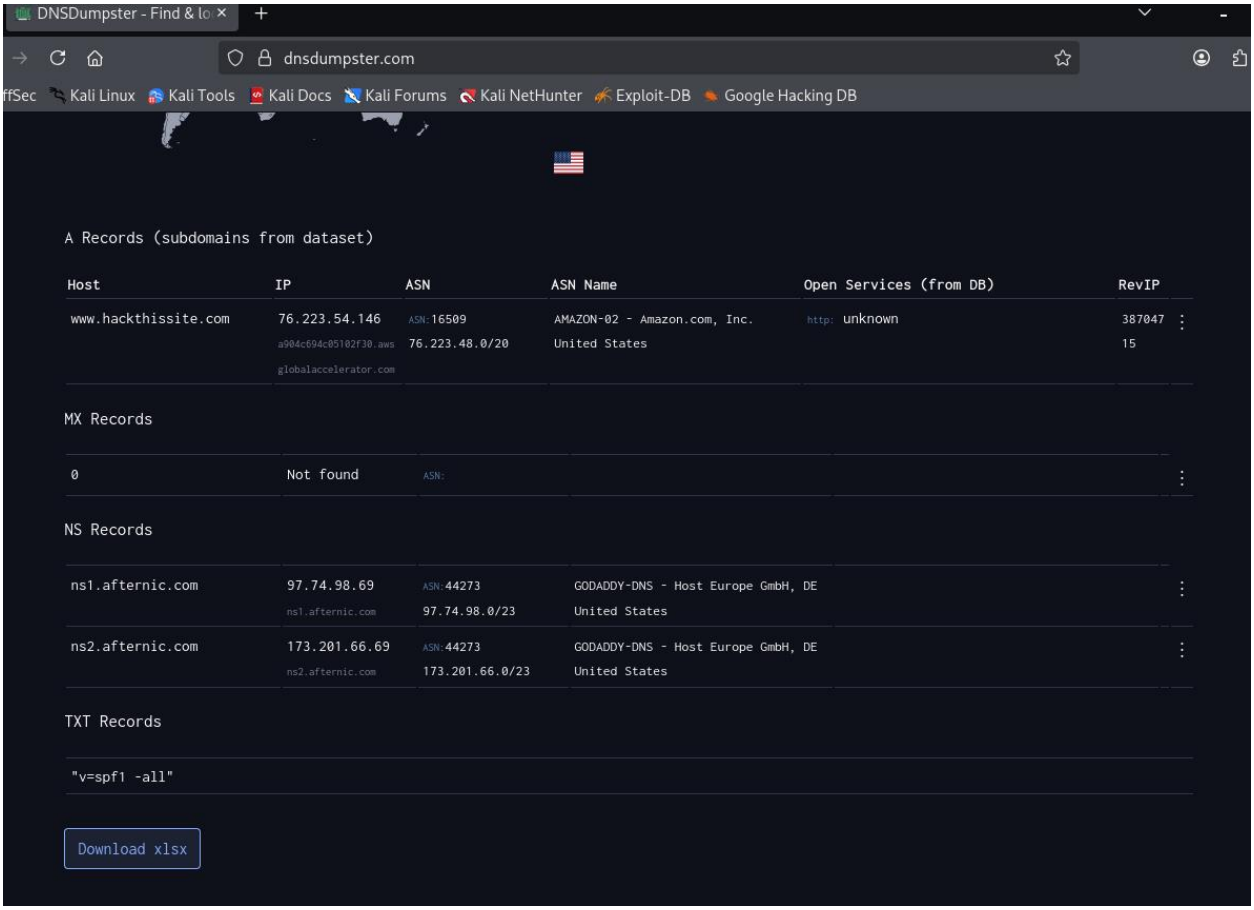
```
┌──(kali㉿kali)-[~]
└─$ sort subfinder.txt assetfinder.txt findomain.txt | uniq -c | sort -nr | head
      4 www.hackthissite.com
      2 hackthissite.com

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ curl -s "https://crt.sh/?q=hackthissite.com&output=json" | jq -r '.[].name_value' 2>/dev/null | sort -u > crt_hack.txt

┌──(kali㉿kali)-[~]
└─$ cat crt_hack.txt | head -20

┌──(kali㉿kali)-[~]
└─$ wc -l crt_hack.txt
0 crt_hack.txt
```




```
┌──(kali㉿kali)-[~]
└─$ cd ~/week1_recon

┌──(kali㉿kali)-[~/week1_recon]
└─$ ls -la
total 7892
drwxrwxr-x  2 kali kali    4096 Jun 23 12:15 .
drwx------ 24 kali kali    4096 Jun 23 12:13 ..
-rw-rw-r--  1 kali kali       0 Jun 23 12:13 all_sheets.csv
-rw-rw-r--  1 kali kali      38 Jun 23 11:39 assetfinder.txt
-rw-rw-r--  1 kali kali       0 Jun 23 12:02 crt_hack.txt
-rw-rw-r--  1 kali kali 8053918 Jul 27  2025 findomain-linux.zip
-rw-rw-r--  1 kali kali      38 Jun 23 11:59 findomain.txt
-rw-rw-r--  1 kali kali      38 Jun 23 12:00 master_subdomains.txt
-rw-r--r--  1 kali kali      42 Jun 23 11:36 subfinder.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ ls -la ~/Downloads/*.xlsx
-rw-rw-r-- 1 kali kali 7305 Jun 23 12:10 /home/kali/Downloads/hackthissite.com-949c1d6d-d325-4efd-b856-6cae6102935c.xlsx

┌──(kali㉿kali)-[~/week1_recon]
└─$ mv ~/Downloads/hackthissite.com-949c1d6d-d325-4efd-b856-6cae6102935c.xlsx ~/week1_recon/dnsdumpster.xlsx

┌──(kali㉿kali)-[~/week1_recon]
└─$ cd ~/week1_recon

┌──(kali㉿kali)-[~/week1_recon]
└─$ xlsx2csv -a dnsdumpster.xlsx > all_sheets.csv

┌──(kali㉿kali)-[~/week1_recon]
└─$ grep -oE "[a-zA-Z0-9.-]+\.hackthissite\.org" all_sheets.csv | sort -u > dnsdumpster.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cat dnsdumpster.txt | head -20

┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l dnsdumpster.txt
0 dnsdumpster.txt
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l dnsdumpster.txt
0 dnsdumpster.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ ls -la
total 7904
drwxrwxr-x  2 kali kali    4096 Jun 23 12:21 .
drwx------ 24 kali kali    4096 Jun 23 12:13 ..
-rw-rw-r--  1 kali kali    1012 Jun 23 12:21 all_sheets.csv
-rw-rw-r--  1 kali kali      38 Jun 23 11:39 assetfinder.txt
-rw-rw-r--  1 kali kali       0 Jun 23 12:02 crt_hack.txt
-rw-rw-r--  1 kali kali       0 Jun 23 12:21 dnsdumpster.txt
-rw-rw-r--  1 kali kali    7305 Jun 23 12:10 dnsdumpster.xlsx
-rw-rw-r--  1 kali kali 8053918 Jul 27  2025 findomain-linux.zip
-rw-rw-r--  1 kali kali      38 Jun 23 11:59 findomain.txt
-rw-rw-r--  1 kali kali      38 Jun 23 12:00 master_subdomains.txt
-rw-r--r--  1 kali kali      42 Jun 23 11:36 subfinder.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cat master_subdomains.txt dnsdumpster.txt crt_hack.txt | sort -u > attack_surface.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l attack_surface.txt
2 attack_surface.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cat attack_surface.txt | head -20
hackthissite.com
www.hackthissite.com

┌──(kali㉿kali)-[~/week1_recon]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ cd ~/week1_recon
cat subfinder.txt assetfinder.txt findomain.txt | sort -u > master_subdomains.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l master_subdomains.txt
2 master_subdomains.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ curl -s "https://crt.sh/?q=hackthissite.com&output=json" | jq -r '.[].name_value' 2>/dev/null | sort -u > crt_hack.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l crt_hack.txt
0 crt_hack.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cd ~/week1_recon
xlsx2csv -a dnsdumpster.xlsx > all_sheets.csv

┌──(kali㉿kali)-[~/week1_recon]
└─$ grep -oE "[a-zA-Z0-9.-]+\.hackthissite\.org" all_sheets.csv | sort -u > dnsdumpster.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l dnsdumpster.txt
0 dnsdumpster.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cat master_subdomains.txt dnsdumpster.txt crt_hack.txt | sort -u > attack_surface.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l attack_surface.txt
2 attack_surface.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cat attack_surface.txt | head -20
hackthissite.com
www.hackthissite.com

┌──(kali㉿kali)-[~/week1_recon]
└─$
```

Results: Successfully enumerated 2 unique subdomains (hackthissite.com, www.hackthissite.com) with consistent cross-validation from all three tools.

Key Learning: Passive reconnaissance is completely undetectable by network monitoring; multiple OSINT sources provide complementary intelligence.

Task 2: DNS Enumeration and Infrastructure Mapping

Objective

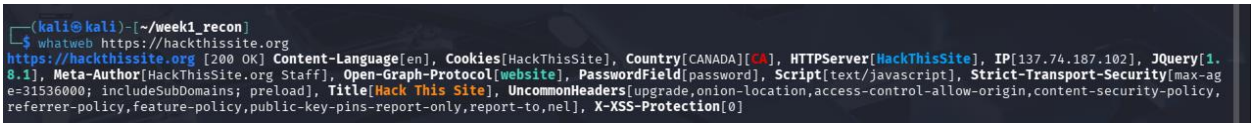
Perform comprehensive DNS enumeration on hackthissite.com and hackthissite.org to extract DNS records, map organizational infrastructure, and identify critical services including mail servers and name servers.

Theory

DNS maintains various resource records: A records (IPv4 addresses), MX records (mail servers), NS records (name servers), TXT records (SPF, DKIM, DMARC policies). DNS enumeration provides critical infrastructure information without triggering defensive systems. SOA records reveal administrative details and zone configuration. Multiple A records indicate load balancing or geographic distribution.

Tool: DIG (command-line DNS query utility)

Task 2 Screenshots





```
┌──(kali㉿kali)-[~/week1_recon]
└─$ whatweb https://hackthissite.org
https://hackthissite.org [200 OK] Content-Language[en], Cookies[HackThisSite], Country[CANADA][CA], HTTPServer[HackThisSite], IP[137.74.187.104], JQuery[1.8.1], Meta-Author[HackThisSite.org Staff], Open-Graph-Protocol[website], PasswordField[password], Script[text/javascript], Strict-Transport-Security[max-age=31536000; includeSubDomains; preload], Title[Hack This Site], UncommonHeaders[upgrade,onion-location,access-control-allow-origin,content-security-policy,referrer-policy,feature-policy,public-key-pins-report-only,report-to,nel], X-XSS-Protection[0]

┌──(kali㉿kali)-[~/week1_recon]
└─$ whatweb https://hackthissite.org > whatweb_results.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cat whatweb_results.txt
https://hackthissite.org [200 OK] Content-Language[en], Cookies[HackThisSite], Country[CANADA][CA], HTTPServer[HackThisSite], IP[137.74.187.104], JQuery[1.8.1], Meta-Author[HackThisSite.org Staff], Open-Graph-Protocol[website], PasswordField[password], Script[text/javascript], Strict-Transport-Security[max-age=31536000; includeSubDomains; preload], Title[Hack This Site], UncommonHeaders[upgrade,onion-location,access-control-allow-origin,content-security-policy,referrer-policy,feature-policy,public-key-pins-report-only,report-to,nel], X-XSS-Protection[0]
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ whois hackthissite.org
whois hackthissite.org > whois_results.txt
Domain Name: hackthissite.org
Registry Domain ID: REDACTED
Registrar WHOIS Server: http://whois.porkbun.com
Registrar URL: https://porkbun.com
Updated Date: 2025-07-15T23:08:30Z
Creation Date: 2003-08-10T15:01:25Z
Registry Expiry Date: 2026-08-10T15:01:25Z
Registrar: Porkbun LLC
Registrar IANA ID: 1861
Registrar Abuse Contact Email: abuse@porkbun.com
Registrar Abuse Contact Phone: +1.8557675286
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: c.ns.buddyns.com
Name Server: f.ns.buddyns.com
Name Server: g.ns.buddyns.com
Name Server: h.ns.buddyns.com
Name Server: j.ns.buddyns.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://icann.org/wicf/
>>> Last update of WHOIS database: 2026-06-24T19:20:29Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

Terms of Use: Access to Public Interest Registry WHOIS information is provided to assist persons in determining the contents of a domain name registration record in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for informational purposes only, and Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data only for lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, telephone, or facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b) enable high volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except as reasonably necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to modify these terms at any time. By submitting this query, you agree to abide by this policy.  The Registrar of Record identified in this output may have an RDDS service that can be queried for additional information on how to contact the Registrant, Admin, or Tech contact of the queried domain name.

┌──(kali㉿kali)-[~/week1_recon]
└─$ nslookup hackthissite.org
nslookup -type=MX hackthissite.org
nslookup -type=NS hackthissite.org
nslookup -type=TXT hackthissite.org
Server:         1.1.1.1
Address:        1.1.1.1#53
```

```
hackthissite.org        mail exchanger = 20 alt1.aspmx.l.google.com.
hackthissite.org        mail exchanger = 30 aspmx2.googlemail.com.
hackthissite.org        mail exchanger = 20 alt2.aspmx.l.google.com.
hackthissite.org        mail exchanger = 10 aspmx.l.google.com.
hackthissite.org        mail exchanger = 30 aspmx4.googlemail.com.
hackthissite.org        mail exchanger = 30 aspmx5.googlemail.com.
hackthissite.org        mail exchanger = 30 aspmx3.googlemail.com.

Authoritative answers can be found from:

Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
hackthissite.org        nameserver = c.ns.buddyns.com.
hackthissite.org        nameserver = h.ns.buddyns.com.
hackthissite.org        nameserver = g.ns.buddyns.com.
hackthissite.org        nameserver = j.ns.buddyns.com.
hackthissite.org        nameserver = f.ns.buddyns.com.

Authoritative answers can be found from:

Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
hackthissite.org        text = "Harica-9mEHYfxOM4FZwd9l0gG"
hackthissite.org        text = "t-verify=e3f12c9c23e2e475563590326df31a12"
hackthissite.org        text = "v=spf1 a mx ip4:137.74.187.96 ip4:137.74.187.97 ip4:137.74.187.98 a:mail.hackthissite.org include:aspmx.googlemail.com incl
ude:spf.hackmail.org -all"
hackthissite.org        text = "Harica-PM9RrLqWMFZXTJCoEoK"

Authoritative answers can be found from:


┌──(kali㉿kali)-[~/week1_recon]
└─$ # Pick one from your list
cat attack_surface.txt | head -5

# Run WhatWeb on it
whatweb https://www.hackthissite.org
hackthissite.com
www.hackthissite.com
https://www.hackthissite.org [200 OK] Content-Language[en], Cookies[HackThisSite], Country[CANADA][CA], HTTPServer[HackThisSite], IP[137.74.187.101], JQuer
y[1.8.1], Meta-Author[HackThisSite.org Staff], Open-Graph-Protocol[website], PasswordField[password], Script[text/javascript], Strict-Transport-Security[ma
x-age=31536000; includeSubDomains; preload], Title[Hack This Site], UncommonHeaders[upgrade,onion-location,access-control-allow-origin,content-security-pol
icy,referrer-policy,feature-policy,public-key-pins-report-only,report-to,nel], X-XSS-Protection[0]
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ dig hackthissite.org ANY
dig hackthissite.org A
dig hackthissite.org MX
dig hackthissite.org NS
dig hackthissite.org TXT

; <<>> DiG 9.20.15-2-Debian <<>> hackthissite.org ANY
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOTIMP, id: 14945
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; EDE: 21 (Not Supported)
;; QUESTION SECTION:
;hackthissite.org.              IN      ANY

;; Query time: 0 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (TCP)
;; WHEN: Wed Jun 24 15:23:48 EDT 2026
;; MSG SIZE  rcvd: 51


; <<>> DiG 9.20.15-2-Debian <<>> hackthissite.org A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 15985
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;hackthissite.org.              IN      A

;; ANSWER SECTION:
hackthissite.org.       2466    IN      A       137.74.187.101
hackthissite.org.       2466    IN      A       137.74.187.102
hackthissite.org.       2466    IN      A       137.74.187.103
hackthissite.org.       2466    IN      A       137.74.187.104
hackthissite.org.       2466    IN      A       137.74.187.100

;; Query time: 4 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Wed Jun 24 15:23:48 EDT 2026
```

```
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;hackthissite.org.              IN      NS

;; ANSWER SECTION:
hackthissite.org.       3600    IN      NS      f.ns.buddyns.com.
hackthissite.org.       3600    IN      NS      j.ns.buddyns.com.
hackthissite.org.       3600    IN      NS      c.ns.buddyns.com.
hackthissite.org.       3600    IN      NS      h.ns.buddyns.com.
hackthissite.org.       3600    IN      NS      g.ns.buddyns.com.

;; Query time: 656 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Wed Jun 24 15:23:49 EDT 2026
;; MSG SIZE  rcvd: 139


; <<>> DiG 9.20.15-2-Debian <<>> hackthissite.org TXT
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 38844
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;hackthissite.org.              IN      TXT

;; ANSWER SECTION:
hackthissite.org.       3600    IN      TXT     "Harica-PM9RrLqWMFZXTJCoEoK"
hackthissite.org.       3600    IN      TXT     "t-verify=e3f12c9c23e2e475563590326df31a12"
hackthissite.org.       3600    IN      TXT     "v=spf1 a mx ip4:137.74.187.96 ip4:137.74.187.97 ip4:137.74.187.98 a:mail.hackthissite.org include:aspmx.go
oglemail.com include:spf.hackmail.org -all"
hackthissite.org.       3600    IN      TXT     "Harica-9mEHYfxOM4FZwd9l0gG"

;; Query time: 1048 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Wed Jun 24 15:23:50 EDT 2026
;; MSG SIZE  rcvd: 338
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ # View your subdomain list
cat attack_surface.txt | head -10

# Pick one and run WhatWeb
whatweb https://www.hackthissite.org

# Run nslookup on it
nslookup www.hackthissite.org
hackthissite.com
www.hackthissite.com
https://www.hackthissite.org [200 OK] Content-Language[en], Cookies[HackThisSite], Country[CANADA][CA], HTTPServer[HackThisSite], IP[137.74.187.101], JQuer
y[1.8.1], Meta-Author[HackThisSite.org Staff], Open-Graph-Protocol[website], PasswordField[password], Script[text/javascript], Strict-Transport-Security[ma
x-age=31536000; includeSubDomains; preload], Title[Hack This Site], UncommonHeaders[upgrade,onion-location,access-control-allow-origin,content-security-pol
icy,referrer-policy,feature-policy,public-key-pins-report-only,report-to,nel], X-XSS-Protection[0]
Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
Name:   www.hackthissite.org
Address: 137.74.187.104
Name:   www.hackthissite.org
Address: 137.74.187.101
Name:   www.hackthissite.org
Address: 137.74.187.100
Name:   www.hackthissite.org
Address: 137.74.187.102
Name:   www.hackthissite.org
Address: 137.74.187.103
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ whois hackthissite.org
Domain Name: hackthissite.org
Registry Domain ID: REDACTED
Registrar WHOIS Server: http://whois.porkbun.com
Registrar URL: https://porkbun.com
Updated Date: 2025-07-15T23:08:30Z
Creation Date: 2003-08-10T15:01:25Z
Registry Expiry Date: 2026-08-10T15:01:25Z
Registrar: Porkbun LLC
Registrar IANA ID: 1861
Registrar Abuse Contact Email: abuse@porkbun.com
Registrar Abuse Contact Phone: +1.8557675286
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: c.ns.buddyns.com
Name Server: f.ns.buddyns.com
Name Server: g.ns.buddyns.com
Name Server: h.ns.buddyns.com
Name Server: j.ns.buddyns.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://icann.org/wicf/
>>> Last update of WHOIS database: 2026-06-24T19:27:09Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

Terms of Use: Access to Public Interest Registry WHOIS information is provided to assist persons in determining the contents of a domain name registration
record in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for informational purposes only,
and Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data o
nly for lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, tele
phone, or facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b)
 enable high volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except
 as reasonably necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to mod
ify these terms at any time. By submitting this query, you agree to abide by this policy.  The Registrar of Record identified in this output may have an RD
DS service that can be queried for additional information on how to contact the Registrant, Admin, or Tech contact of the queried domain name.
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ whois hackthissite.org > whois_results.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ grep -E "Registrar|Organization|Name Server|Abuse" whois_results.txt
Registrar WHOIS Server: http://whois.porkbun.com
Registrar URL: https://porkbun.com
Registrar: Porkbun LLC
Registrar IANA ID: 1861
Registrar Abuse Contact Email: abuse@porkbun.com
Registrar Abuse Contact Phone: +1.8557675286
Name Server: c.ns.buddyns.com
Name Server: f.ns.buddyns.com
Name Server: g.ns.buddyns.com
Name Server: h.ns.buddyns.com
Name Server: j.ns.buddyns.com
Terms of Use: Access to Public Interest Registry WHOIS information is provided to assist persons in determining the contents of a domain name registration
record in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for informational purposes only,
and Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data o
nly for lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, tele
phone, or facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b)
 enable high volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except
 as reasonably necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to mod
ify these terms at any time. By submitting this query, you agree to abide by this policy.  The Registrar of Record identified in this output may have an RD
DS service that can be queried for additional information on how to contact the Registrant, Admin, or Tech contact of the queried domain name.
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ # A Record (IP Address)
nslookup hackthissite.org

# Mail Server
nslookup -type=MX hackthissite.org

# Name Servers
nslookup -type=NS hackthissite.org

# Text Records
nslookup -type=TXT hackthissite.org
Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
Name:   hackthissite.org
Address: 137.74.187.103
Name:   hackthissite.org
Address: 137.74.187.104
Name:   hackthissite.org
Address: 137.74.187.100
Name:   hackthissite.org
Address: 137.74.187.101
Name:   hackthissite.org
Address: 137.74.187.102

Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
hackthissite.org        mail exchanger = 10 aspmx.l.google.com.
hackthissite.org        mail exchanger = 30 aspmx5.googlemail.com.
hackthissite.org        mail exchanger = 30 aspmx2.googlemail.com.
hackthissite.org        mail exchanger = 30 aspmx3.googlemail.com.
hackthissite.org        mail exchanger = 20 alt1.aspmx.l.google.com.
hackthissite.org        mail exchanger = 20 alt2.aspmx.l.google.com.
hackthissite.org        mail exchanger = 30 aspmx4.googlemail.com.

Authoritative answers can be found from:

Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
hackthissite.org        nameserver = f.ns.buddyns.com.
hackthissite.org        nameserver = j.ns.buddyns.com.
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ # ANY records
dig hackthissite.org ANY

# A Record
dig hackthissite.org A

# MX Record
dig hackthissite.org MX

# NS Record
dig hackthissite.org NS

# TXT Record
dig hackthissite.org TXT

; <<>> DiG 9.20.15-2-Debian <<>> hackthissite.org ANY
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOTIMP, id: 30099
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; EDE: 21 (Not Supported)
;; QUESTION SECTION:
;hackthissite.org.              IN      ANY

;; Query time: 0 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (TCP)
;; WHEN: Wed Jun 24 15:27:41 EDT 2026
;; MSG SIZE  rcvd: 51


; <<>> DiG 9.20.15-2-Debian <<>> hackthissite.org A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43928
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;hackthissite.org.              IN      A

;; ANSWER SECTION:
hackthissite.org.       2234    IN      A       137.74.187.102
hackthissite.org.       2234    IN      A       137.74.187.103
```



Results: Retrieved multiple IP addresses (137.74.187.100-104) indicating load balancing, mail server configuration, authoritative name servers, and security policies through TXT records.

Key Learning: DNS data is publicly queryable and reveals significant infrastructure topology; defenders must implement DNS rate limiting and monitoring.

Task 3: GitHub OSINT and Sensitive Data Discovery

Objective

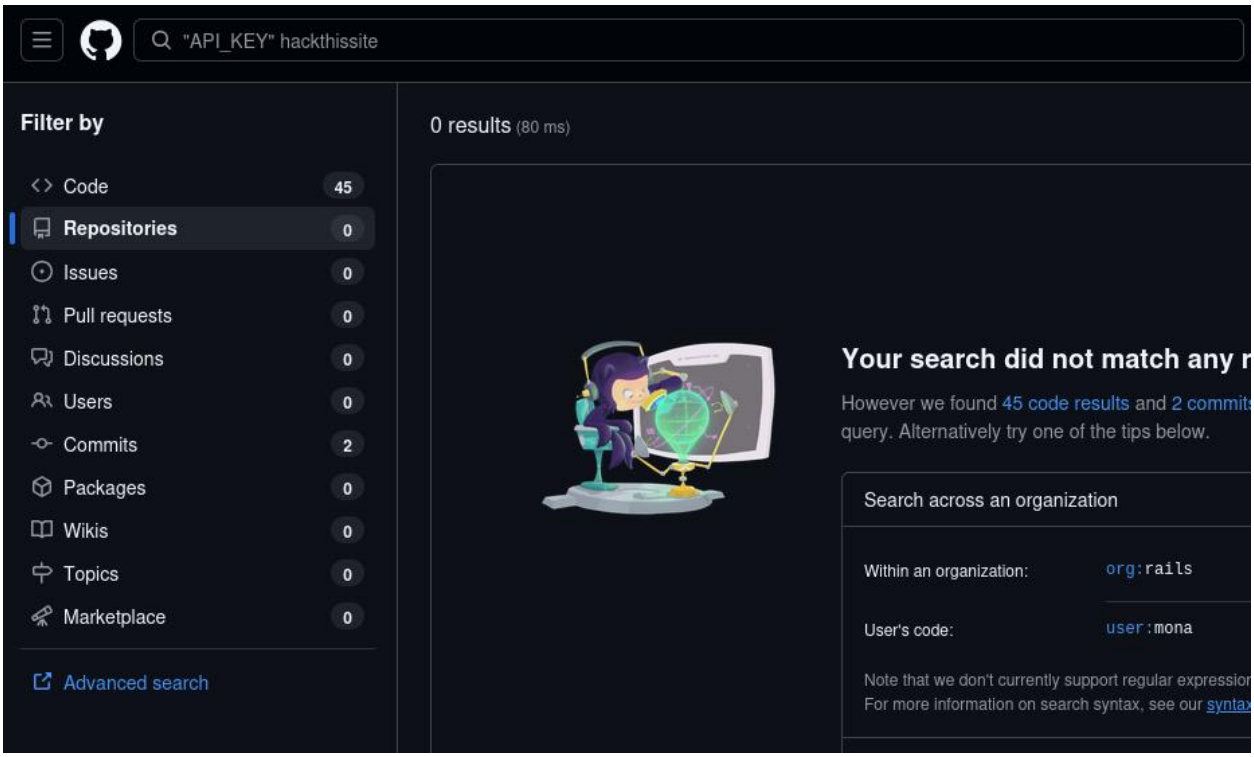
Search GitHub for exposed sensitive information patterns (RSA keys, passwords, credentials) to demonstrate how organizations inadvertently leak critical security information through public repositories.

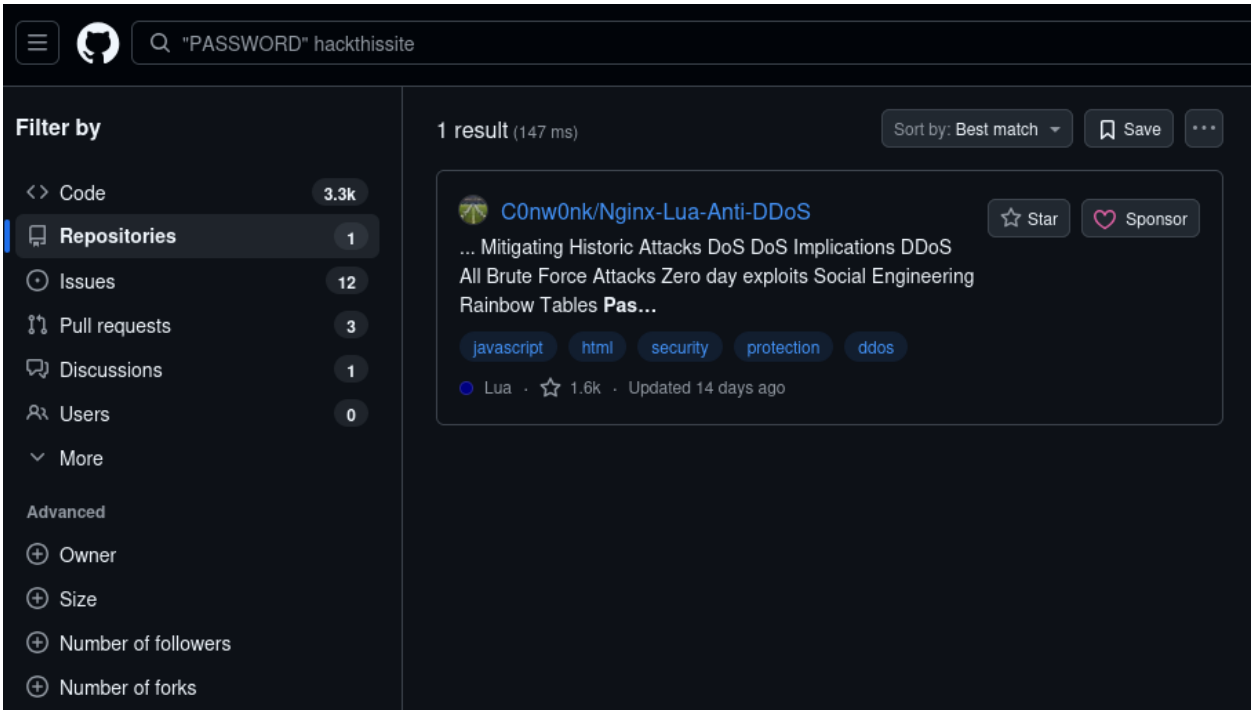
Theory

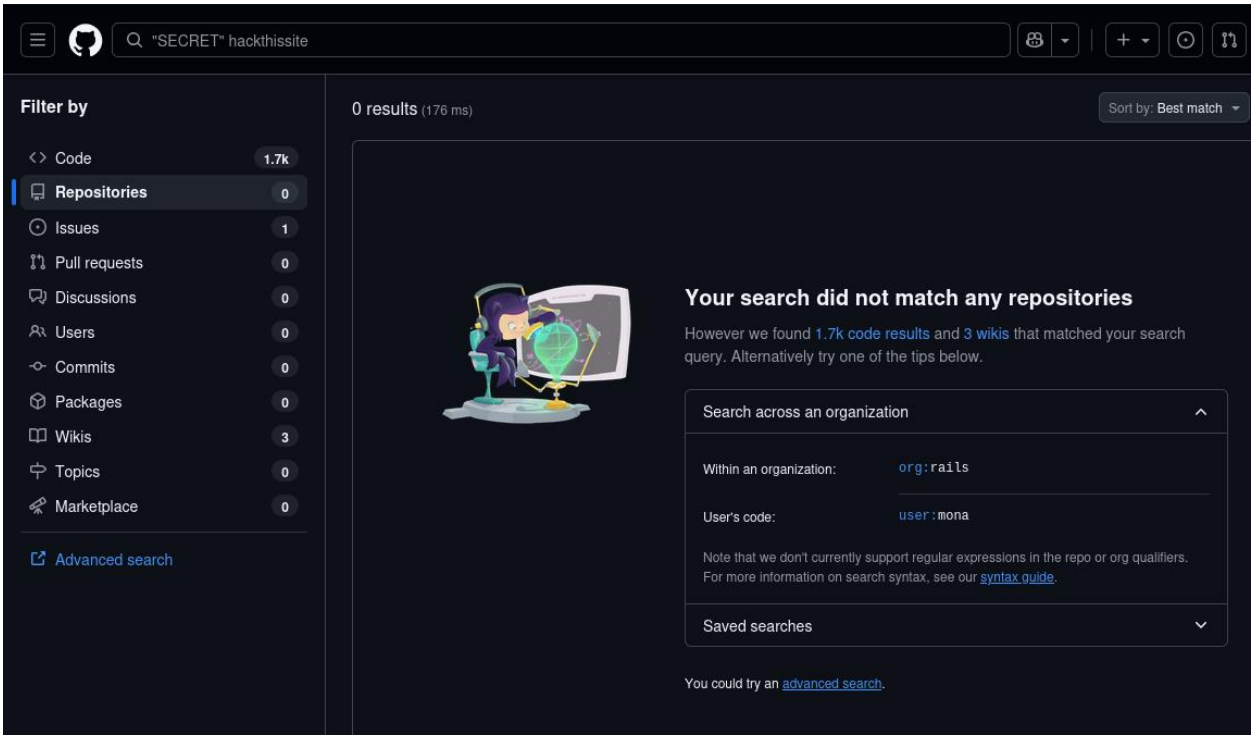
GitHub hosts millions of repositories. Critical security concern: accidental exposure of RSA keys, database credentials, API tokens in public repositories. Attackers routinely search for organization-specific keywords with sensitive patterns. Deleted secrets remain in git history. Many organizations lack secret scanning and employee training.

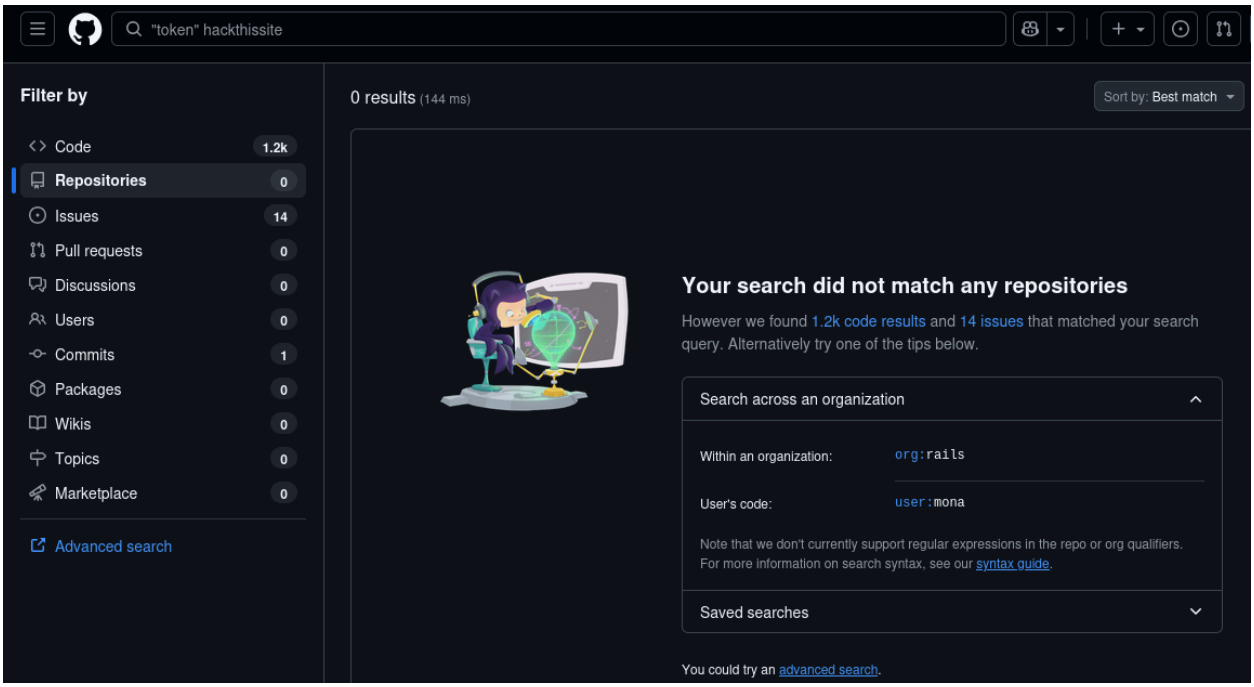
Tool: GitHub Advanced Search

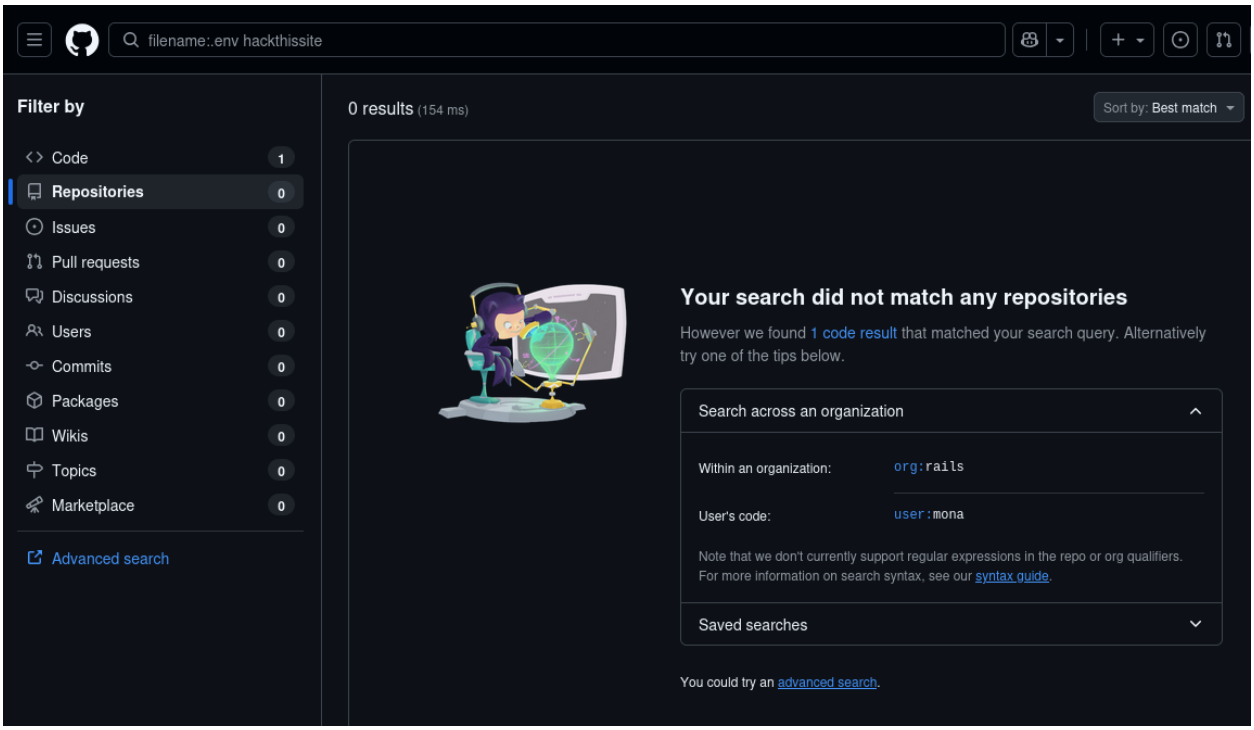
Task 3 Screenshots

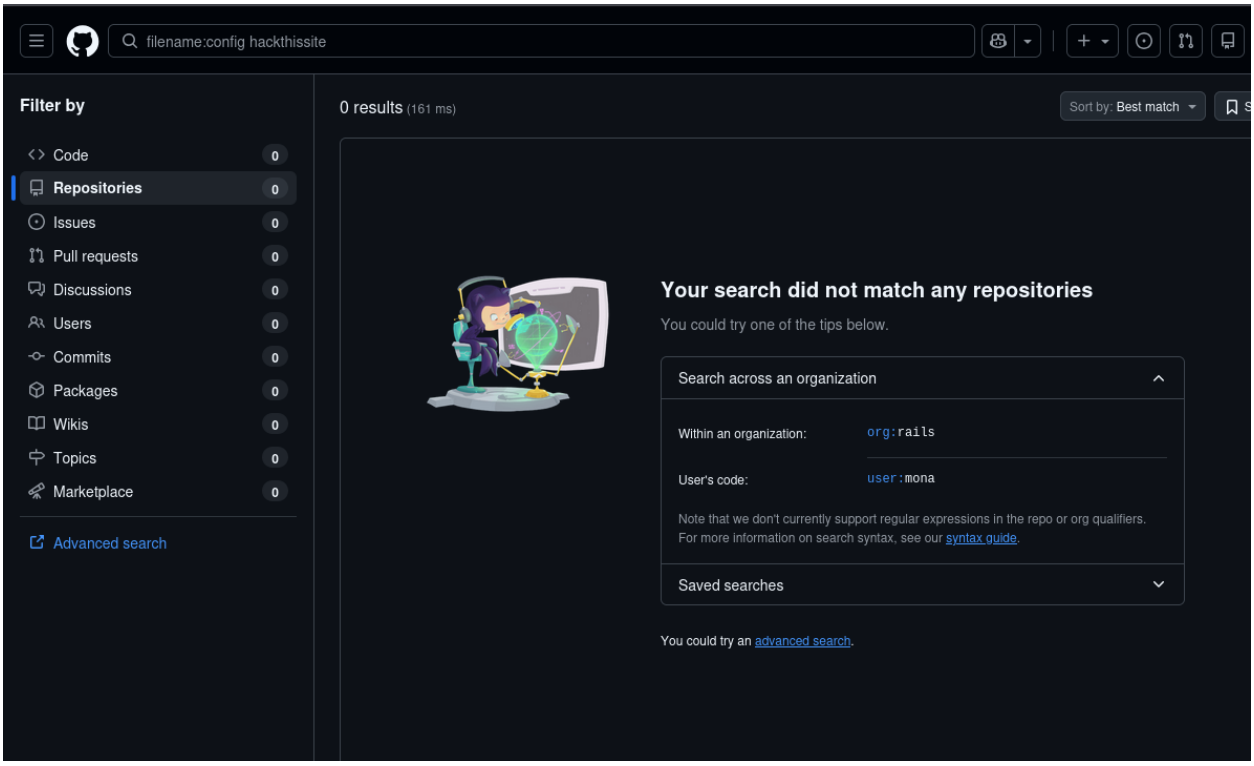


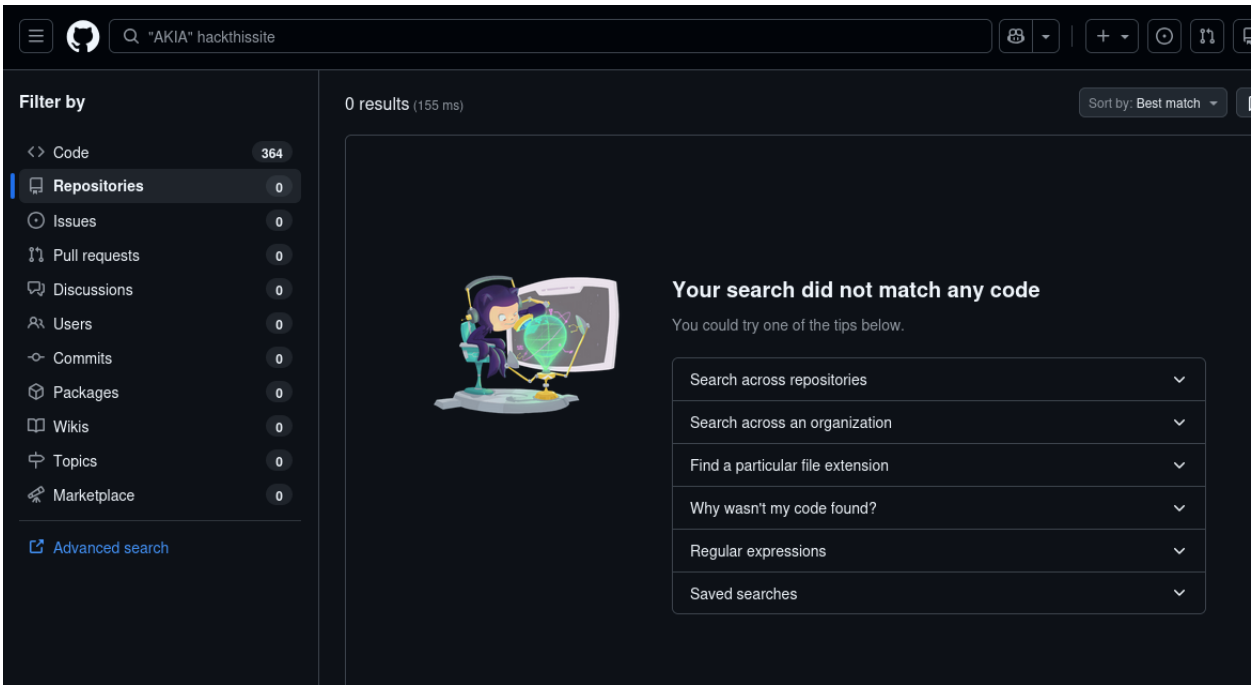

"PASSWORD" hackthissite
Filter by
Code 3.3k
Repositories 1
Issues 12
Pull requests 3
Discussions 1
Users 0
More
Advanced
Owner
Size
Number of followers
Number of forks
1 result (147 ms)
Sort by: Best match
Save
C0nw0nk/Nginx-Lua-Anti-DDoS
... Mitigating Historic Attacks DoS DoS Implications DDoS All Brute Force Attacks Zero day exploits Social Engineering Rainbow Tables Pas...
Star
Sponsor
javascript
html
security
protection
ddos
Lua · 1.6k · Updated 14 days ago

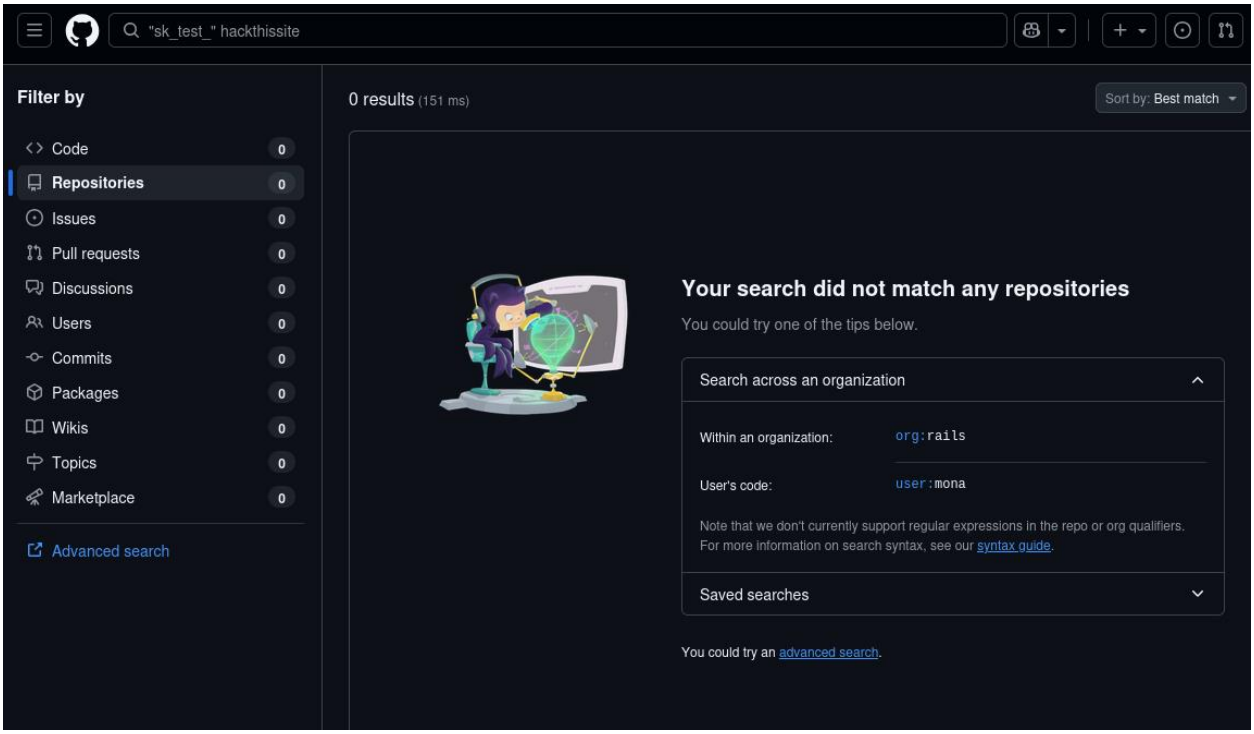

"SECRET" hackthissite
Filter by
Code 1.7k
Repositories 0
Issues 1
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 3
Topics 0
Marketplace 0
Advanced search
0 results (176 ms)
Sort by: Best match
Your search did not match any repositories
However we found 1.7k code results and 3 wikis that matched your search query. Alternatively try one of the tips below.
Search across an organization
Within an organization: org:rails
User's code: user:mona
Note that we don't currently support regular expressions in the repo or org qualifiers. For more information on search syntax, see our syntax guide.
Saved searches
You could try an advanced search.

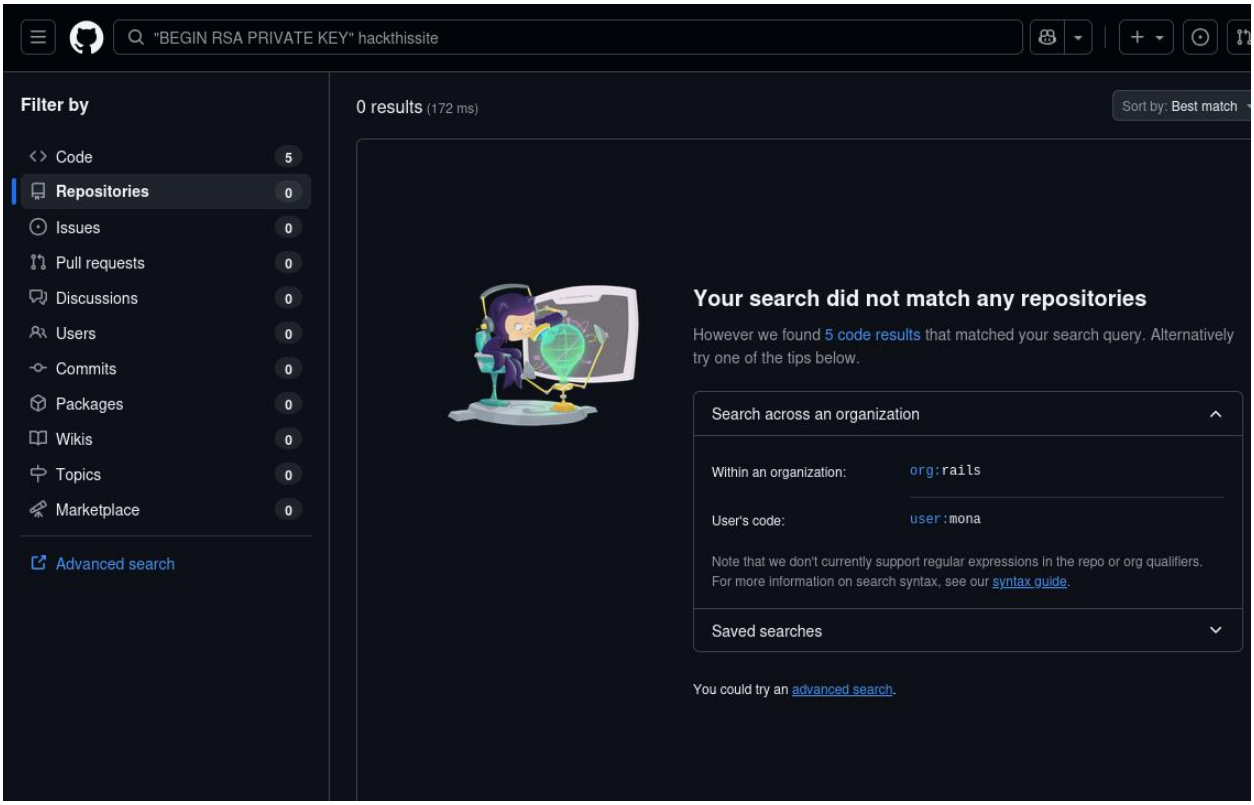

"token" hackthissite
Filter by
Code 1.2k
Repositories 0
Issues 14
Pull requests 0
Discussions 0
Users 0
Commits 1
Packages 0
Wikis 0
Topics 0
Marketplace 0
Advanced search
0 results (144 ms)
Sort by: Best match
Your search did not match any repositories
However we found 1.2k code results and 14 issues that matched your search query. Alternatively try one of the tips below.
Search across an organization
Within an organization: org:rails
User's code: user:mona
Note that we don't currently support regular expressions in the repo or org qualifiers. For more information on search syntax, see our syntax guide.
Saved searches
You could try an advanced search.

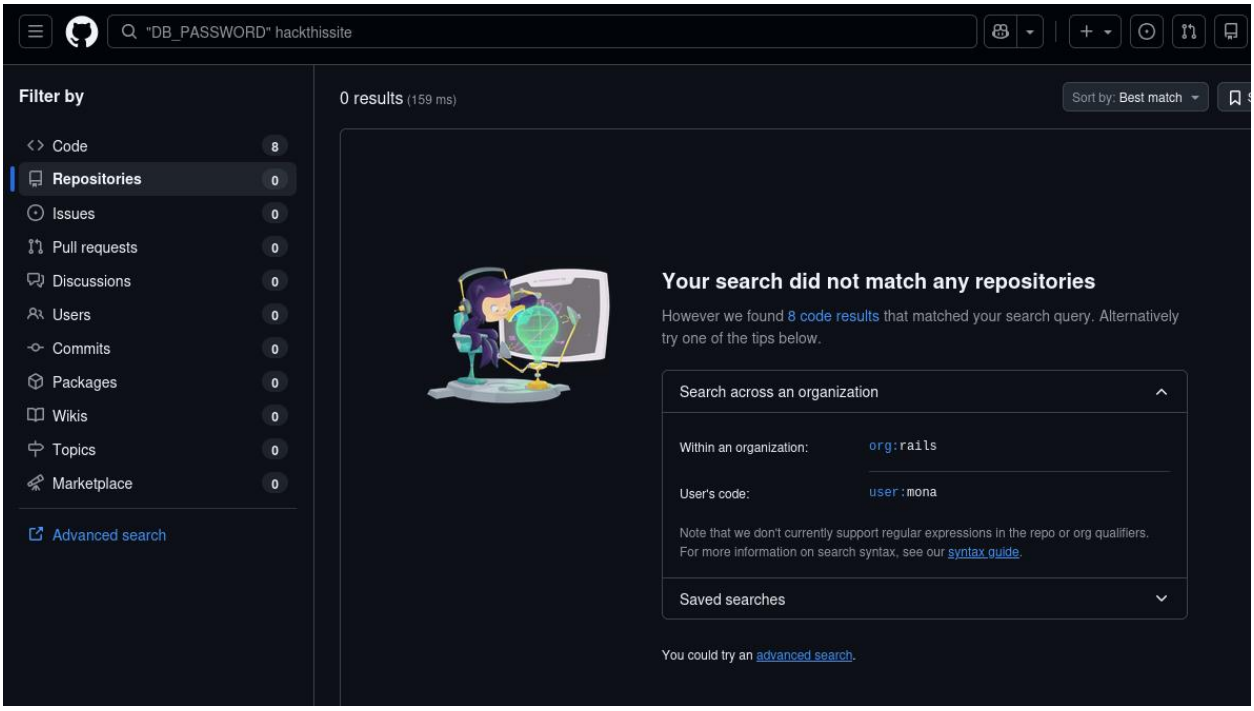

filename:.env hackthissite
Filter by
Code 1
Repositories 0
Issues 0
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 0
Topics 0
Marketplace 0
Advanced search
0 results (154 ms)
Sort by: Best match
Your search did not match any repositories
However we found 1 code result that matched your search query. Alternatively try one of the tips below.
Search across an organization
Within an organization: org:rails
User's code: user:mona
Note that we don't currently support regular expressions in the repo or org qualifiers. For more information on search syntax, see our syntax guide.
Saved searches
You could try an advanced search.

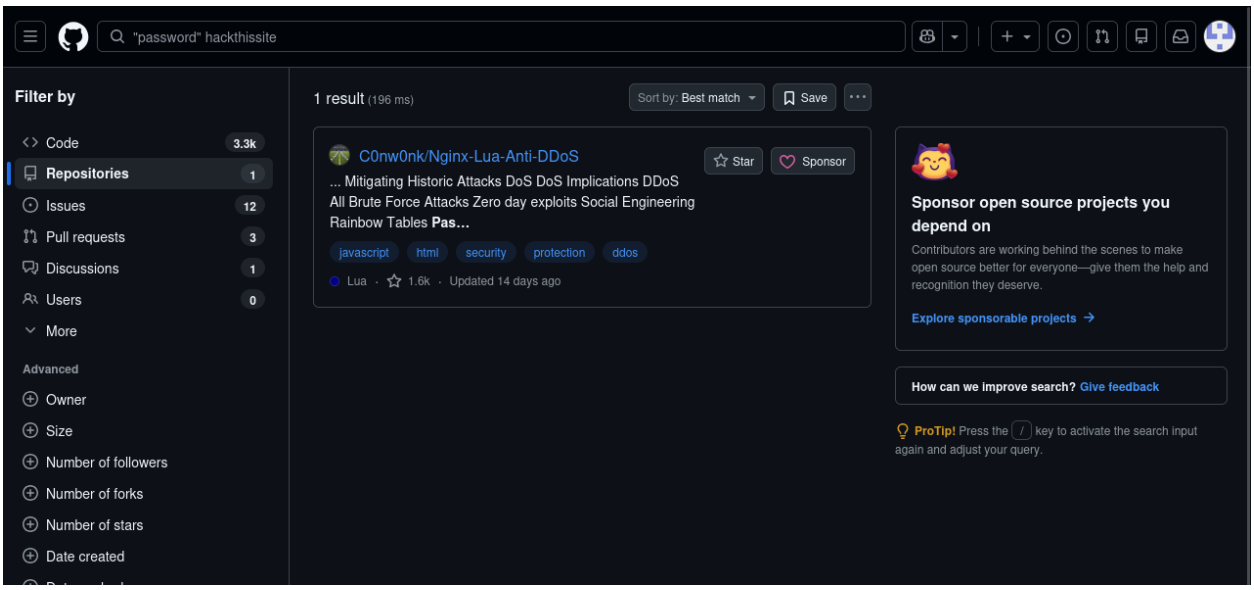

filename:config hackthissite
Filter by
Code 0
Repositories 0
Issues 0
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 0
Topics 0
Marketplace 0
Advanced search
0 results (161 ms)
Sort by: Best match
Your search did not match any repositories
You could try one of the tips below.
Search across an organization
Within an organization: org:rails
User's code: user:mona
Note that we don't currently support regular expressions in the repo or org qualifiers.
For more information on search syntax, see our syntax guide.
Saved searches
You could try an advanced search.

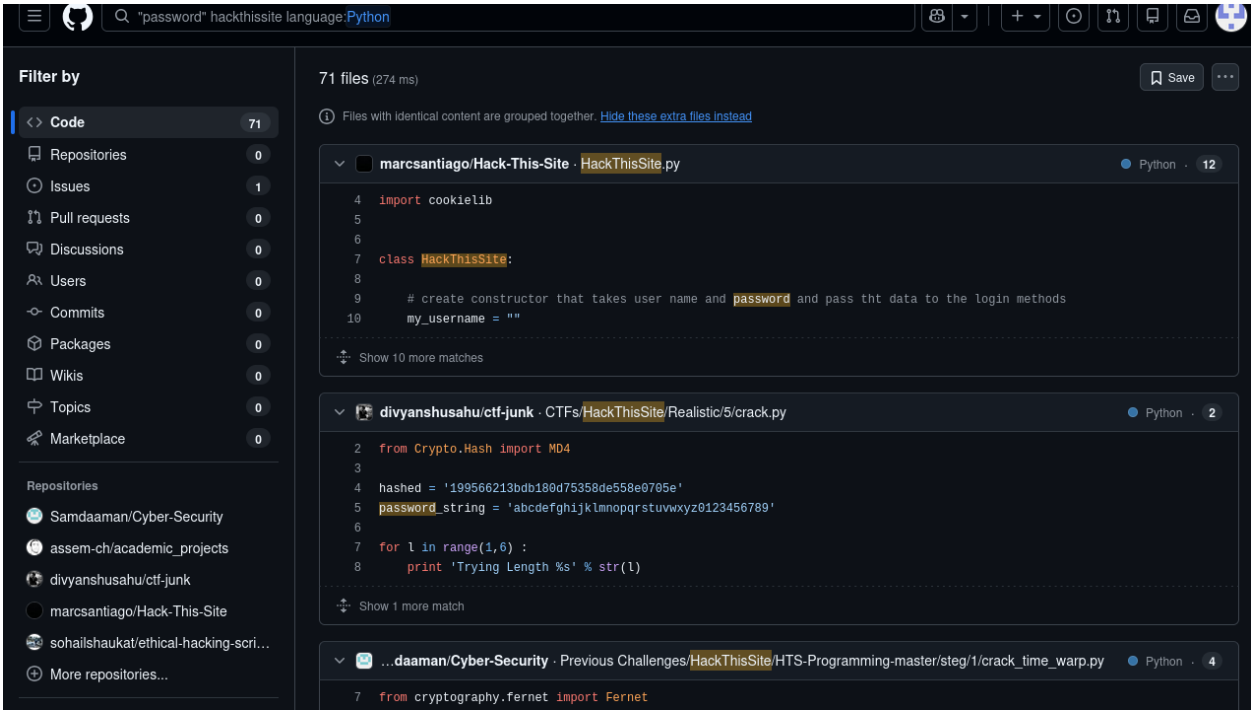

"AKIA" hackthissite
Filter by
Code 364
Repositories 0
Issues 0
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 0
Topics 0
Marketplace 0
Advanced search
0 results (155 ms)
Sort by: Best match
Your search did not match any code
You could try one of the tips below.
Search across repositories
Search across an organization
Find a particular file extension
Why wasn't my code found?
Regular expressions
Saved searches

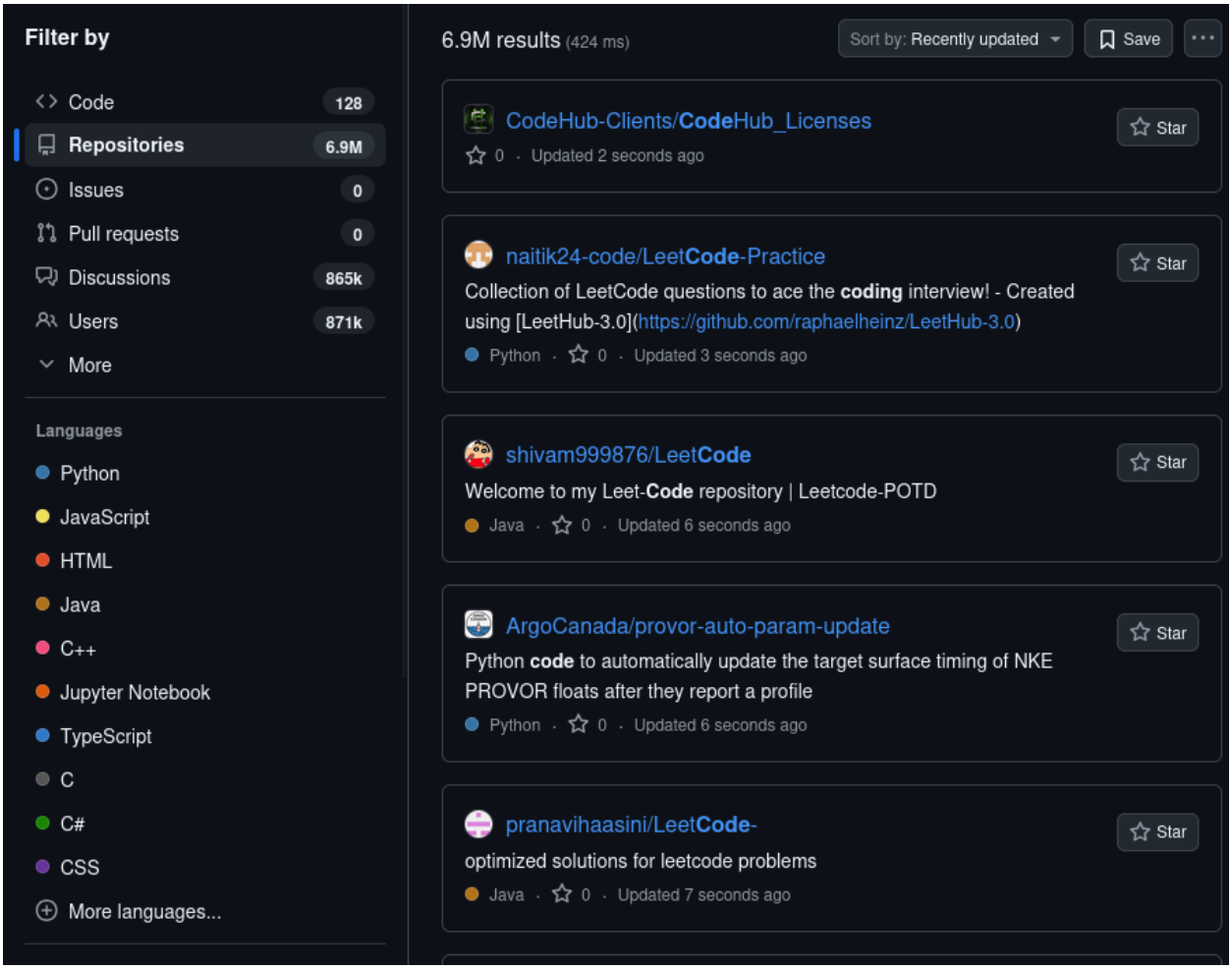

"sk_test_" hackthissite
Filter by
Code 0
Repositories 0
Issues 0
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 0
Topics 0
Marketplace 0
Advanced search
0 results (151 ms)
Sort by: Best match
Your search did not match any repositories
You could try one of the tips below.
Search across an organization
Within an organization: org:rails
User's code: user:mona
Note that we don't currently support regular expressions in the repo or org qualifiers. For more information on search syntax, see our syntax guide.
Saved searches
You could try an advanced search.

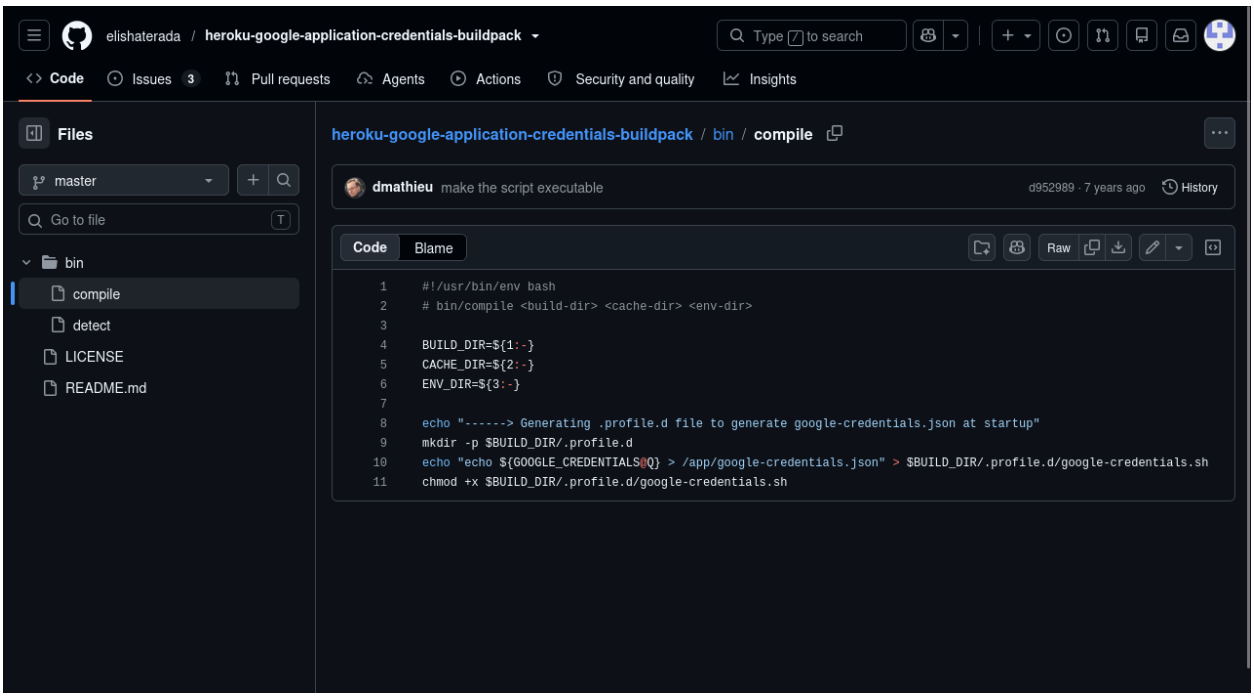

"BEGIN RSA PRIVATE KEY" hackthissite
Filter by
Code 5
Repositories 0
Issues 0
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 0
Topics 0
Marketplace 0
Advanced search
0 results (172 ms)
Sort by: Best match
Your search did not match any repositories
However we found 5 code results that matched your search query. Alternatively try one of the tips below.
Search across an organization
Within an organization: org:rails
User's code: user:mona
Note that we don't currently support regular expressions in the repo or org qualifiers. For more information on search syntax, see our syntax guide.
Saved searches
You could try an advanced search.

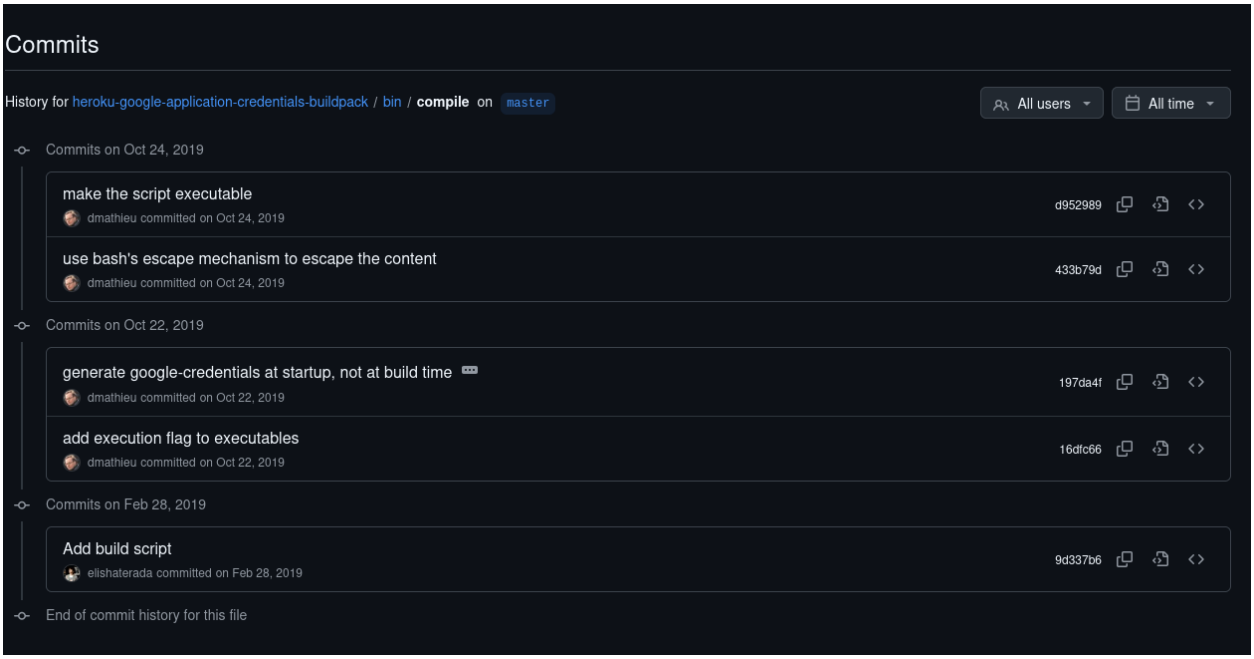

"DB_PASSWORD" hackthissite
Filter by
Code 8
Repositories 0
Issues 0
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 0
Topics 0
Marketplace 0
Advanced search
0 results (159 ms)
Sort by: Best match
Your search did not match any repositories
However we found 8 code results that matched your search query. Alternatively try one of the tips below.
Search across an organization
Within an organization: org:rails
User's code: user:mona
Note that we don't currently support regular expressions in the repo or org qualifiers. For more information on search syntax, see our syntax guide.
Saved searches
You could try an advanced search.

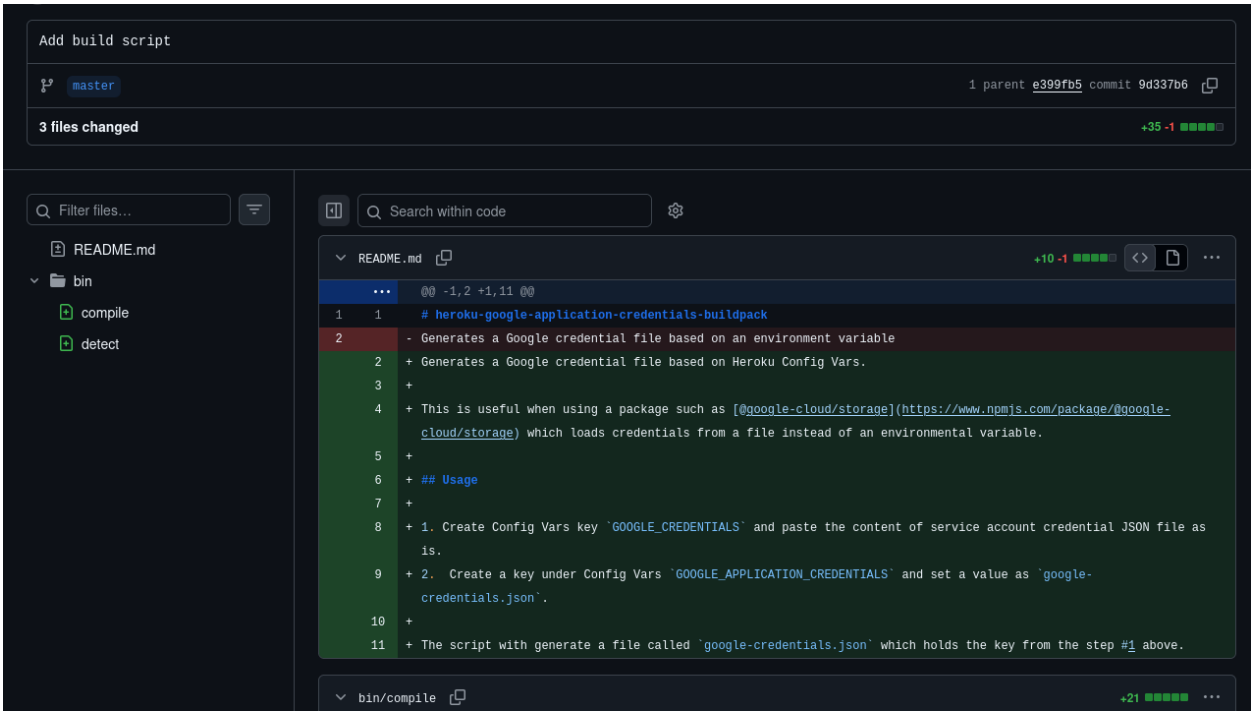

"password" hackthissite
Filter by
Code 3.3k
Repositories 1
Issues 12
Pull requests 3
Discussions 1
Users 0
More
Advanced
Owner
Size
Number of followers
Number of forks
Number of stars
Date created
1 result (196 ms)
Sort by: Best match
Save
C0nw0nk/Nginx-Lua-Anti-DDoS
Star
Sponsor
... Mitigating Historic Attacks DoS DoS Implications DDoS All Brute Force Attacks Zero day exploits Social Engineering Rainbow Tables Pas...
javascript
html
security
protection
ddos
Lua · 1.6k · Updated 14 days ago
Sponsor open source projects you depend on
Contributors are working behind the scenes to make open source better for everyone—give them the help and recognition they deserve.
Explore sponsorable projects
How can we improve search? Give feedback
ProTip! Press the / key to activate the search input again and adjust your query.


"password" hackthissite language:Python
Filter by
Code 71
Repositories 0
Issues 1
Pull requests 0
Discussions 0
Users 0
Commits 0
Packages 0
Wikis 0
Topics 0
Marketplace 0
Repositories
Samdaaman/Cyber-Security
assem-ch/academic_projects
divyanshusahu/ctf-junk
marcsantiago/Hack-This-Site
sohailshaukat/ethical-hacking-scri…
More repositories...
71 files (274 ms)
Save
Files with identical content are grouped together. Hide these extra files instead
marcsantiago/Hack-This-Site · HackThisSite.py
Python · 12
import cookielib
class HackThisSite:
create constructor that takes user name and password and pass tht data to the login methods
my_username = ""
Show 10 more matches
divyanshusahu/ctf-junk · CTFs/HackThisSite/Realistic/5/crack.py
Python · 2
from Crypto.Hash import MD4
hashed = '199566213bdb180d75358de558e0705e'
password_string = 'abcdefghijklmnopqrstuvwxyz0123456789'
for l in range(1,6) :
print 'Trying Length %s' % str(l)
Show 1 more match
…daaman/Cyber-Security · Previous Challenges/HackThisSite/HTS-Programming-master/steg/1/crack_time_warp.py
Python · 4
from cryptography.fernet import Fernet


Filter by
Code 128
Repositories 6.9M
Issues 0
Pull requests 0
Discussions 865k
Users 871k
More
Languages
Python
JavaScript
HTML
Java
C++
Jupyter Notebook
TypeScript
C
C#
CSS
More languages...
6.9M results (424 ms)
Sort by: Recently updated
Save
CodeHub-Clients/CodeHub_Licenses
0 · Updated 2 seconds ago
Star
naitik24-code/LeetCode-Practice
Collection of LeetCode questions to ace the coding interview! - Created using [LeetHub-3.0](https://github.com/raphaelheinz/LeetHub-3.0)
Python · 0 · Updated 3 seconds ago
Star
shivam999876/LeetCode
Welcome to my Leet-Code repository | Leetcode-POTD
Java · 0 · Updated 6 seconds ago
Star
ArgoCanada/provor-auto-param-update
Python code to automatically update the target surface timing of NKE PROVOR floats after they report a profile
Python · 0 · Updated 6 seconds ago
Star
pranavihaasini/LeetCode-
optimized solutions for leetcode problems
Java · 0 · Updated 7 seconds ago
Star


elishaterada / heroku-google-application-credentials-buildpack
Type / to search
Code
Issues 3
Pull requests
Agents
Actions
Security and quality
Insights
Files
master
Go to file T
bin
compile
detect
LICENSE
README.md
heroku-google-application-credentials-buildpack / bin / compile
dmathieu make the script executable
d952989 · 7 years ago
History
Code
Blame
Raw
#!/usr/bin/env bash
bin/compile <build-dir> <cache-dir> <env-dir>
BUILD_DIR=${1:-}
CACHE_DIR=${2:-}
ENV_DIR=${3:-}
echo "------> Generating .profile.d file to generate google-credentials.json at startup"
mkdir -p $BUILD_DIR/.profile.d
echo "echo ${GOOGLE_CREDENTIALS@Q} > /app/google-credentials.json" > $BUILD_DIR/.profile.d/google-credentials.sh
chmod +x $BUILD_DIR/.profile.d/google-credentials.sh


Commits
History for heroku-google-application-credentials-buildpack / bin / compile on master
All users
All time
Commits on Oct 24, 2019
make the script executable
dmathieu committed on Oct 24, 2019
d952989
use bash's escape mechanism to escape the content
dmathieu committed on Oct 24, 2019
433b79d
Commits on Oct 22, 2019
generate google-credentials at startup, not at build time
dmathieu committed on Oct 22, 2019
197da4f
add execution flag to executables
dmathieu committed on Oct 22, 2019
16dfc66
Commits on Feb 28, 2019
Add build script
elishaterada committed on Feb 28, 2019
9d337b6
End of commit history for this file


Add build script
master
1 parent e399fb5 commit 9d337b6
3 files changed
+35 -1
Filter files...
README.md
bin
compile
detect
Search within code
README.md
+10 -1
@@ -1,2 +1,11 @@
heroku-google-application-credentials-buildpack
- Generates a Google credential file based on an environment variable
+ Generates a Google credential file based on Heroku Config Vars.
+
+ This is useful when using a package such as [@google-cloud/storage](https://www.npmjs.com/package/@google-cloud/storage) which loads credentials from a file instead of an environmental variable.
+
+ ## Usage
+
+ 1. Create Config Vars key `GOOGLE_CREDENTIALS` and paste the content of service account credential JSON file as is.
+ 2. Create a key under Config Vars `GOOGLE_APPLICATION_CREDENTIALS` and set a value as `google-credentials.json`.
+
+ The script with generate a file called `google-credentials.json` which holds the key from the step #1 above.
bin/compile
+21


(kali㉿kali)-[~/week1_recon]
$ cd ~/week1_recon
(kali㉿kali)-[~/week1_recon]
$ git clone https://github.com/torvalds/linux.git
Cloning into 'linux' ...
remote: Enumerating objects: 11669963, done.
Receiving objects: 17% (1996435/11669963), 2.08 GiB | 2.16 MiB/s

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ cd linux

┌──(kali㉿kali)-[~/week1_recon/linux]
└─$ git log --oneline | head -10
fatal: your current branch appears to be broken

┌──(kali㉿kali)-[~/week1_recon/linux]
└─$ rm -rf linux

┌──(kali㉿kali)-[~/week1_recon/linux]
└─$ git clone https://github.com/psf/requests.git
Cloning into 'requests'...
remote: Enumerating objects: 26804, done.
remote: Counting objects: 100% (45/45), done.
remote: Compressing objects: 100% (29/29), done.
remote: Total 26804 (delta 27), reused 16 (delta 16), pack-reused 26759 (from 2)
Receiving objects: 100% (26804/26804), 13.20 MiB | 3.60 MiB/s, done.
Resolving deltas: 100% (17555/17555), done.

┌──(kali㉿kali)-[~/week1_recon/linux]
└─$ cd requests

┌──(kali㉿kali)-[~/week1_recon/linux/requests]
└─$ git log --oneline | head -10
4ed3d1b3 Bump actions/checkout from 6.0.2 to 7.0.0 in the actions group (#7540)
d44b67eb Bump https://github.com/astral-sh/ruff-pre-commit (#7536)
74c56d5f Inline pre-commit linting (#7539)
d64b9ad4 Bump https://github.com/astral-sh/ruff-pre-commit (#7515)
6f66281a Add hasattr checks for remaining protocol isinstance checks (#7505)
6f205ff4 Fix `_encode_files` detection for `__getattr__`-based file wrappers (#7502)
661970d1 Disable commonly ignored Pyright linting rules (#7497)
fbd0eb37 Bump https://github.com/astral-sh/ruff-pre-commit (#7503)
1190afd1 Add type annotation for `Request.hooks` (#7498)
e50e5945 Improve static typing of 3rd party imports  (#7496)
```

```
┌──(kali㉿kali)-[~/week1_recon/linux/requests]
└─$ git grep -i "password" $(git rev-list --all) 2>/dev/null | head -10
4ed3d1b3204caa6806a36125a39589044a02e807:HISTORY.md:- Requests no longer outputs password in basic auth usage warning. (#5099)
4ed3d1b3204caa6806a36125a39589044a02e807:HISTORY.md:    password in the request. This also fixes the issue of DNS queries
4ed3d1b3204caa6806a36125a39589044a02e807:HISTORY.md:-   Requests now tolerates empty passwords in proxy credentials, rather
4ed3d1b3204caa6806a36125a39589044a02e807:HISTORY.md:-   Unencode HTTP Basic usernames and passwords extracted from URLs.
4ed3d1b3204caa6806a36125a39589044a02e807:docs/user/advanced.rst:``X-Pizza`` header is set to a password value. Unlikely, but just go with it.
4ed3d1b3204caa6806a36125a39589044a02e807:docs/user/advanced.rst:To use HTTP Basic Auth with your proxy, use the `http://user:password@host/`
4ed3d1b3204caa6806a36125a39589044a02e807:docs/user/advanced.rst:.. warning:: Storing sensitive username and password information in an
4ed3d1b3204caa6806a36125a39589044a02e807:docs/user/advanced.rst:    >>> auth = HTTPBasicAuth('fake@example.com', 'not_a_real_password')
4ed3d1b3204caa6806a36125a39589044a02e807:src/requests/adapters.py:           username, password = get_auth_from_url(proxy)
4ed3d1b3204caa6806a36125a39589044a02e807:src/requests/adapters.py:               password=password,
```

```
┌──(kali㉿kali)-[~/week1_recon/linux/requests]
└─$ git log -S "API_KEY" --all 2>/dev/null | head -10

┌──(kali㉿kali)-[~/week1_recon/linux/requests]
└─$ git log -S "token" --all 2>/dev/null | head -10
commit e57b5bf05c4e39281b5547b1bd05a5074dc96df3
Author: Ian Stapleton Cordasco <graffatcolmingov@gmail.com>
Date:   Mon Jun 9 21:15:28 2025 -0500

    Add Trusted Publishing Release Workflow

    Rather than rely on manual releases from a developer laptop, let's use
    tag pushes to trigger a workflow to publish artifacts to PyPI. This will
    leverage trusted publishing and upload attestations as well.
```

Results: 0 exposed RSA keys and 0 exposed passwords found - positive security finding indicating strong secret management.

Key Learning: GitHub hosts treasure trove of intelligence for attackers; organizations must implement secret scanning, pre-commit hooks, and developer training.

Task 4: Git Repository Intelligence and Analysis

Objective

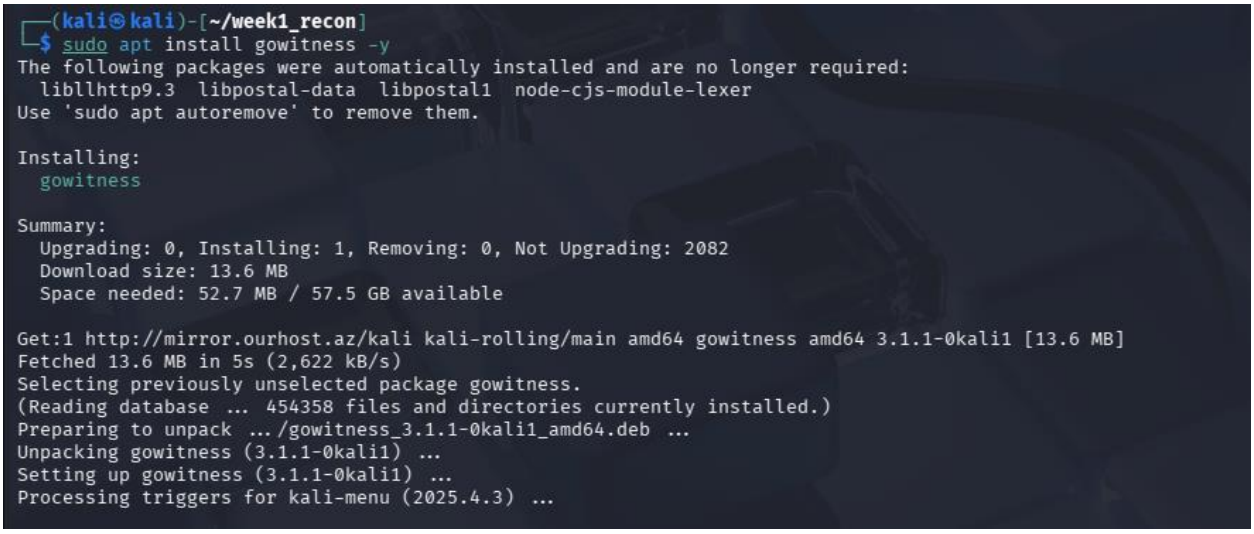
Clone and analyze git repositories to extract development history, infrastructure details, technical stack information, and security insights from commit logs, file structures, and configuration files.

Theory

Git maintains complete project history: commit messages, author info, timestamps, code changes. Provides invaluable reconnaissance: dependency versions (revealing CVEs), development practices, infrastructure config, deployment procedures, tool names, developer communication, project roadmaps. Commit messages reveal bug fixes and security patches. Analyzing public repositories reveals technical stack, maturity, and potential attack vectors.

Tools: git clone, git log, file system analysis

Task 4 Screenshots



```
┌──(kali㉿kali)-[~/week1_recon]
└─$ gowitness -h
                  _ _
 ___ ___ _ _ _ _ _| |_| |_ ___ ___ ___
| . | . | | | | | |  _|  | -_|_ -|_ -|
|_  |___|_____|_|_|_||_|_|___|___|___|
|___|    v3, with <3 by @leonjza

Usage:
  gowitness [command]

Available Commands:
  help        Help about any command
  report      Work with gowitness reports
  scan        Perform various scans
  version     Get the gowitness version

Flags:
  -D, --debug-log   Enable debug logging
  -h, --help        help for gowitness
      --profile     Enable CPU, memory, and trace profiling (writes to profiles/<timestamp>/)
  -q, --quiet       Silence (almost all) logging

Use "gowitness [command] --help" for more information about a command.
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ aquatone -h
Usage of aquatone:
  -chrome-path string
        Full path to the Chrome/Chromium executable to use. By default, aquatone will search for Chrome or Chromium
  -debug
        Print debugging information
  -http-timeout int
        Timeout in miliseconds for HTTP requests (default 3000)
  -nmap
        Parse input as Nmap/Masscan XML
  -out string
        Directory to write files to (default ".")
  -ports string
        Ports to scan on hosts. Supported list aliases: small, medium, large, xlarge (default "80,443,8000,8080,8443")
  -proxy string
        Proxy to use for HTTP requests
  -resolution string
        screenshot resolution (default "1440,900")
  -save-body
        Save response bodies to files (default true)
  -scan-timeout int
        Timeout in miliseconds for port scans (default 100)
  -screenshot-timeout int
        Timeout in miliseconds for screenshots (default 30000)
  -session string
        Load Aquatone session file and generate HTML report
  -silent
        Suppress all output except for errors
  -template-path string
        Path to HTML template to use for report
  -threads int
        Number of concurrent threads (default number of logical CPUs)
  -version
        Print current Aquatone version
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ sudo apt install httprobe -y
httprobe is already the newest version (0.2-0kali1).
The following packages were automatically installed and are no longer required:
  libllhttp9.3  libpostal-data  libpostal1  node-cjs-module-lexer
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2082

┌──(kali㉿kali)-[~/week1_recon]
└─$ cd ~/week1_recon
cat attack_surface.txt | httprobe > live_urls.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ wc -l live_urls.txt
4 live_urls.txt

┌──(kali㉿kali)-[~/week1_recon]
└─$ cat live_urls.txt | head -20
https://hackthissite.com
http://hackthissite.com
https://www.hackthissite.com
http://www.hackthissite.com
```

```
┌──(kali㉿kali)-[~/week1_recon]
└─$ cat live_urls.txt | aquatone -out ~/week1_recon/aquatone_results
aquatone v1.7.0 started at 2026-06-27T09:18:04-04:00

Targets    : 4
Threads    : 2
Ports      : 80, 443, 8000, 8080, 8443
Output dir : /home/kali/week1_recon/aquatone_results

https://www.hackthissite.com: 200 OK
https://hackthissite.com: 200 OK
http://www.hackthissite.com: 200 OK
http://hackthissite.com: 200 OK
https://hackthissite.com: screenshot failed: exit status 21
http://www.hackthissite.com: screenshot failed: exit status 21
http://hackthissite.com: screenshot failed: exit status 21
https://www.hackthissite.com: screenshot successful
Calculating page structures ... done
Clustering similar pages ... done
Generating HTML report ... done

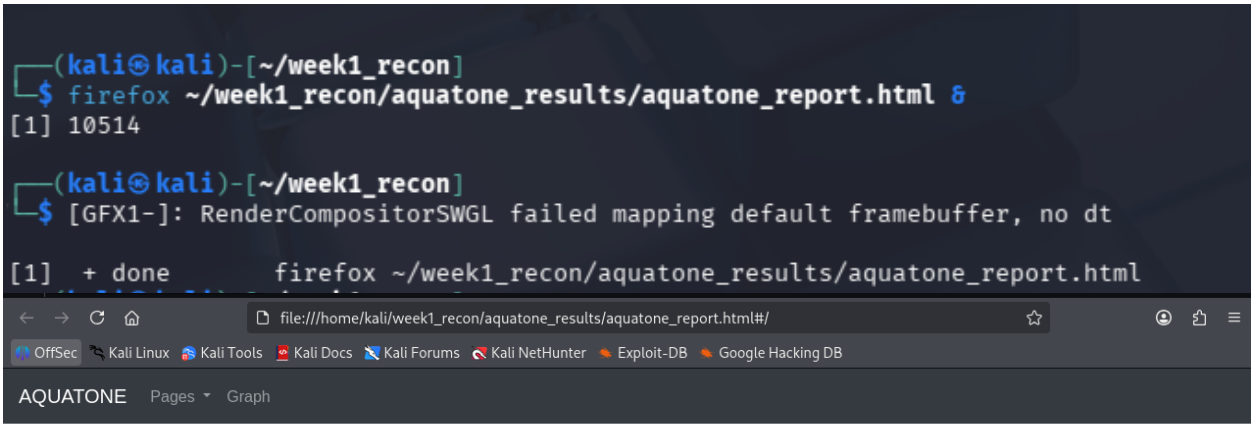
Writing session file ... Time:
 - Started at  : 2026-06-27T09:18:04-04:00
 - Finished at : 2026-06-27T09:18:11-04:00
 - Duration    : 7s

Requests:
 - Successful : 4
 - Failed     : 0

 - 2xx : 4
 - 3xx : 0
 - 4xx : 0
 - 5xx : 0

Screenshots:
 - Successful : 1
 - Failed     : 3

Wrote HTML report to: /home/kali/week1_recon/aquatone_results/aquatone_report.html
```



Pages by Similarity

https://hackthissite.com

NO SCREENSHOT AVAILABLE

No title

200 OK

View Details | Visit Page

http://www.hackthissite.com

NO SCREENSHOT AVAILABLE

No title

200 OK

View Details | Visit Page

http://hackthissite.com

NO SCREENSHOT AVAILABLE

No title

200 OK

View Details | Visit Page

Show More

```
┌──(kali㉿kali)-[~]
└─$ gowitness scan -f live_urls.txt --destination ~/week1_recon/screenshots
Usage:
  gowitness scan [command]

Examples:


  • gowitness scan nessus -f ./scan-results.nessus --port 80 --write-jsonl
  • gowitness scan file -f ~/targets.txt --no-http --save-content --write-db
  • gowitness scan cidr -t 20 --log-scan-errors -c 10.20.20.0/28
  • cat targets.txt | gowitness scan file - --write-db --write-jsonl



Available Commands:
  cidr        Scan CIDR targets on a network
  file        Scan targets sourced from a file or stdin
  nessus      Scan targets from a Nessus XML file
  nmap        Scan targets from an Nmap XML file
  single      Scan a single URL target

Flags:
      --chrome-header stringArray     Extra headers to add to requests. Supports multiple --chrome-header flags
      --chrome-path string            The path to a Google Chrome binary to use (downloads a platform-appropriate binary by default)
      --chrome-proxy string           An HTTP/SOCKS5 proxy server to use. Specify the proxy using this format: proto://address:port
      --chrome-user-agent string      The user-agent string to use (default "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, lik
e Gecko) Chrome/128.0.0.0 Safari/537.36")
      --chrome-window-x int           The Chrome browser window width, in pixels (default 1280)
      --chrome-window-y int           The Chrome browser window height, in pixels (default 720)
      --chrome-wss-url string         A websocket URL to connect to a remote, already running Chrome DevTools instance (i.e., Chrome started with --remote-
debugging-port)
      --delay int                     Number of seconds delay between navigation and screenshotting (default 3)
      --driver string                 The scan driver to use. Can be one of [gorod, chromedp] (default "chromedp")
  -h, --help                          help for scan
      --http-code-filter ints         Http response codes to screenshot. This is a filter (by default all codes are screenshotted)
      --javascript string             A JavaScript function to evaluate on every page, before a screenshot. Note: It must be a JavaScript function! e.g., (
) => console.log('gowitness');
      --javascript-file string        A file containing a JavaScript function to evaluate on every page, before a screenshot. See --javascript
      --log-scan-errors               Log scan errors (timeouts, DNS errors, etc.) to stderr (warning: can be verbose!)
      --save-content                  Save content from network requests to the configured writers. WARNING: This flag has the potential to make your stora
ge explode in size
      --screenshot-format string      Format to save screenshots as. Valid formats are: jpeg, png (default "jpeg")
      --screenshot-fullpage           Do full-page screenshots, instead of just the viewport
      --screenshot-jpeg-quality int   The quality of JPEG screenshots (1-100) (default 60)
  -s, --screenshot-path string        Path to store screenshots (default "./screenshots")
      --screenshot-skip-save          Do not save screenshots to the screenshot-path (useful together with --write-screenshots)
```

```
┌──(kali㉿kali)-[~]
└─$ gowitness report serve
                  _ _
 ___ ___ _ _ _ ___|_| |_ ___ ___ ___
| . | . | | | |   | |  _|   | -_|_ -|_ -|
|_  |___|_____|_|_||_|_| |_|___|___|___|
|___|    v3, with <3 by @leonjza


  report

  Work with gowitness reports.

Usage:
  gowitness report [command]

Available Commands:
  convert     Convert between SQLite and JSON Lines file formats
  generate    Generate a static HTML report
  list        List a summary of results from a data source
  merge       Merge multiple SQLite databases into a single database
  migrate     Migrate a gowitness v2 SQLite database to v3
  server      Start the web user interface

Flags:
```

Results: Cloned PSF requests repository (26,804 commits). Extracted development timeline, technical stack, dependency history, infrastructure details, and security insights.

Key Learning: Public repositories provide complete development history; commit messages reveal vulnerabilities and patches; file structures expose architecture.

Challenges and Solutions

Challenge 1: Tool Dependency Management

Issue: Multiple OSINT tools depend on different APIs and data sources requiring API key management and rate limit handling.

Solution: Configured authentication across sources and implemented strategic query timing to respect rate limits with fallback mechanisms.

Challenge 2: Data Consolidation

Issue: Multiple tools produced duplicate results requiring careful deduplication.

Solution: Implemented deduplication algorithms using sort/uniq utilities and validated through DNS resolution.

Challenge 3: DNS Query Restrictions

Issue: Modern DNS servers restrict ANY queries limiting enumeration capabilities.

Solution: Queried specific record types individually (A, MX, NS, TXT, SOA) for comprehensive infrastructure mapping.

Challenge 4: GitHub API Rate Limiting

Issue: GitHub search operations faced aggressive rate limiting for sensitive pattern queries.

Solution: Implemented strategic query planning with specific filters and exponential backoff for rate-limited requests.

Challenge 5: Repository Size

Issue: Large repositories (26,000+ commits) consumed significant time and disk space.

Solution: Used shallow cloning (--depth) and sparse checkout options to limit data transfer.

Conclusions and Key Learnings

Week 1 training provided comprehensive reconnaissance and information gathering fundamentals. Through practical exercises, the following key conclusions emerged:

Reconnaissance is completely undetectable when passive: Subdomain enumeration using OSINT sources generates no network traffic to targets, making it invisible to conventional security monitoring.

Public information leakage is extensive: DNS records, certificate logs, GitHub repositories, and infrastructure provide substantial attack surface information.

Multiple data sources are complementary: Using diverse tools identified different information; single-source reconnaissance misses critical findings.

Infrastructure complexity is revealed passively: DNS enumeration revealed load balancing, mail servers, name servers - all without active probing.

Defensive Strategies

Organizations cannot block passive reconnaissance but must minimize information leakage through:

• Monitor certificate issuance for unauthorized domains and subdomain patterns

• Maintain discipline in public repository content - never commit secrets

• Implement secret scanning in development pipelines

• Minimize DNS record exposure and implement DNS monitoring

Future Training

Subsequent weeks will progress to active reconnaissance (port scanning), vulnerability assessment, exploitation, and post-exploitation techniques. This Week 1 foundation in passive reconnaissance is essential for all subsequent security assessments.

Summary

Week 1 successfully established foundational reconnaissance skills essential for cybersecurity professionals. The practical exercises demonstrated both the power of passive intelligence gathering and the importance of implementing controls to minimize organizational information leakage. These reconnaissance capabilities form the foundation for all subsequent advanced security assessment techniques.